



From perception to reaction: an RLCC-engine-driven cybersecurity framework for smart agricultural ecosystems

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Abstract

Agriculture 4.0 makes significant use of interconnected sensors, autonomous farm equipment, cloud-based farm management systems, and data-driven decision-making. Even though these tools make farm operations more productive and sustainable, they also expose farm equipment, Internet of Things (IoT) devices, supply-chain management systems, and farm data to Cybersecurity attacks. Cybersecurity incidents in the agricultural sector may result in operational disruption, financial loss, and the compromise of data sovereignty. To address these risks, this study adopts a three-stage analytical approach consisting of challenge identification through literature and infrastructure analysis, prioritisation of the identified challenges, and systematic mapping onto a four-layer Agricultural Data-sharing Ecosystem (ADE) architecture comprising perception, network, edge, and cloud/application layers. The analysis identifies six structural challenges driving Cybersecurity risks in agriculture: data management, data governance, integration and interoperability, data privacy and security, knowledge gaps, and infrastructure limitations. Based on these findings, a conceptual multi-layer Cybersecurity framework is proposed that introduces a dedicated *Reaction Layer* enabling coordinated prevention, detection, and mitigation across ADE layers. The operational logic of the *Reaction Layer* is formalised through the Reaction Layer Cross-Layer Coordination (RLCC) algorithm, which specifies the sequential coordination of prevention, detection, mitigation, and audit actions across the ADE infrastructure layers, and links each response action to the identified challenges. To support the transition from conceptual design to practical deployment, an initial prototype architecture is outlined. It identifies which components can use existing technologies and which require new development. It also defines the scenarios and metrics that will be used to evaluate the framework.

Keywords Agriculture 4.0 · Cybersecurity · Agricultural data-sharing ecosystem (ADE) · *Reaction layer* · Cross-layer security · RLCC algorithm

1 Introduction

While the land remains fixed [48], according to United Nations (UN) projections, the global population may reach nine billion by the middle of the century. This represents an important challenge to the farming industry, leading to an increased demand of almost 70% in the food production rate and competition and exploitation of natural resources [38, 41]. According to the Food and Agriculture Organization of the United Nations, from 2017, we are living in the fourth agricultural generation (Agriculture 4.0). This period emphasizes the integration of digital technologies and data-driven farming. [38]. Therefore, Agriculture 4.0 represents more than just a movement [58]. It involves the use of sensors, Internet of Things (IoT) devices, remote control tools, and other cutting-edge technologies such as cobots (collaborative robots) and AI (Artificial Intelligence) to give farmers

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increased management and awareness of their fields and operations [29], provide facilities for informed decisions, and improve integration with supply chain partners [38].

The digital transformation of agriculture has significantly expanded the sector's cyber-attack surface, exposing data-driven farming systems, automation platforms, and agri-food supply chains to cyber threats. In recent years, ransomware attacks have increasingly targeted the agriculture and food sectors, causing operational disruptions, financial losses, and cascading impacts on food supply chains. Reports from the Federal Bureau of Investigation and the European Union Agency for Cybersecurity confirm that information technology and operational technology systems in agriculture are frequently exploited by cybercriminals. These impacts affect organizations ranging from small farms to large producers and processors [34, 35]. Media-reported incidents, such as attacks on robotic milking systems and food processing facilities, further demonstrate that cybersecurity threats in Agriculture 4.0 are concrete and operationally disruptive [27, 59]. A recent review of cybersecurity in smart farming also found that denial-of-service (DoS) attacks on smart farming services and malware attacks on digital systems are the most common threats. The review explains that these threats are mainly caused by weak security practices, outdated software, and a lack of regulations designed for the agriculture sector [50]. These developments highlight the need for an integrated cybersecurity architecture that coordinates threat prevention, detection, and response strategies across the Agricultural Data-sharing Ecosystem (ADE).

ADE refers to the complex network of data exchanges, digital platforms, and stakeholders that interact within modern agricultural settings. Previous studies describe ADE as an integrated structure that connects data sources, infrastructures, and actors involved in digital farming systems [18, 23, 41, 62]. Rather than introducing ADE as a new concept, this study models it as a layered architectural framework to support systematic cybersecurity analysis. Following established smart agriculture and Internet of Things (IoT) designs, the ADE is modelled across the perception, network, edge, and cloud/application layers. This abstraction captures cross-layer dependencies and security requirements, providing the structural foundation for analyzing cybersecurity challenges and developing the proposed framework. Motivated by the increasing frequency and impact of Cybersecurity incidents in Agriculture 4.0, as well as the limitations of existing fragmented security approaches, this study makes several key contributions:

- First, it presents a structured analysis of Cybersecurity challenges in data-driven agriculture by synthesizing technical, governance-related, and human-centric perspectives. Through this analysis, the study identifies

and prioritizes six core challenges (CH1–CH6) using a literature-driven methodology.

- Second, it proposes a novel multi-layer Cybersecurity framework for ADEs that explicitly integrates the perception, network, edge, and cloud/application layers.
- Third, the coordination logic of the Reaction Layer is formalized through the RLCC algorithm, which specifies how prevention, detection, mitigation, and audit actions are sequentially coordinated across ADE infrastructure layers, with each step directly traceable to challenges CH1–CH6.
- Finally, an initial prototype architecture is outlined, identifying which components can be realised using existing technologies and which require development from scratch, providing a concrete path toward practical deployment.

By explicitly linking real-world Cybersecurity risks to a structured architectural model, the proposed framework provides a basis for coordinated security enforcement across the layers of an Agriculture 4.0 ecosystem. The rest of this paper is structured as follows.

Section 2 reviews related work on Cybersecurity in Agriculture 4.0, data governance, and cloud and edge security. Section 3 describes the methodology used to identify and prioritise Cybersecurity challenges and to derive the proposed framework. Section 4 analyses the structural challenges driving Cybersecurity risks in Agriculture 4.0, emphasising the barriers that must be addressed to build stakeholder trust and promote acceptance of Agrifood 4.0 technologies. Section 5 discusses the transition from traditional data management toward data governance in ADEs. Section 6 introduces the ADE architecture and outlines its layered structure. Section 7 analyses interconnectivity and security issues arising from heterogeneous agricultural infrastructures. Section 8 presents the proposed conceptual framework, the role of the *Reaction Layer* in enabling coordinated cross-layer responses, and the RLCC algorithm that formalises its operational logic. Section 9 outlines an initial prototype architecture and the metrics by which the framework would be validated. Section 10 outlines future research directions, and Sect. 11 concludes the paper.

2 Related work

This section provides an overview of previous research on Cybersecurity in Agriculture 4.0 and ADEs. Rather than providing a paper-by-paper summary, the review is organized thematically to highlight dominant research directions, their strengths, and the limitations that motivate the need for a holistic, cross-layer Cybersecurity framework.

Table 1 Comparison of cybersecurity approaches in Agriculture 4.0 according to ADE layers

Study	Focus area	Per.	Net.	Ed.	Cl.	Broad challenges	RL	CLC	FA
[41]	IoT security	✓	✓	×	×	Data security and Privacy	×	×	×
[36]	Intrusion detection	✓	✓	×	×	Data security and Privacy	×	×	×
[30]	IoT architecture	✓	✓	×	×	Integration; Security	×	×	×
[67]	Secure communication	×	✓	×	×	Data security and Privacy	×	×	×
[16]	Authentication scheme	✓	✓	×	×	Data security and Privacy	×	×	×
[11]	Data governance	×	×	×	✓	Data management; Governance	×	×	×
[5]	Compliance & governance	×	×	×	✓	Data Governance	×	×	×
[23]	Trust frameworks	×	×	×	✓	Data Governance; Knowledge gaps	×	×	×
[31]	Data governance	×	×	×	✓	Data Governance	×	×	×
[21]	Data transparency	×	×	×	✓	Data Governance	×	×	×
[8]	Cloud/Edge security	×	×	✓	✓	Security; Integration	×	×	×
[4]	Cloud protection	×	×	×	✓	Security	×	×	×
[10]	Access control	×	×	×	✓	Security	×	×	×
[32]	Cyberbiosecurity	✓	×	×	✓	Multiple security aspects	×	×	×
[15]	Blockchain security	✓	✓	×	✓	Data management; Security	×	×	×
[22]	Cyber threat intelligence (CTI)	✓	×	×	✓	Security	×	×	×
[7]	Security testbed (dairy)	✓	✓	×	✓	Data security; Infrastructure (Empirical)	×	×	×
[45]	Threats & farmer protection	✓	✓	×	✓	Security; Knowledge gaps; Governance (Review)	×	×	×
[24]	AI-IoT-Cybersecurity	✓	✓	✓	✓	Integration; Security; Governance (Conceptual)	×	×	×
[50]	Threats & mitigation review	✓	✓	✓	✓	Security; Knowledge gaps; Governance (Review)	×	×	×
This study	Multi-layered Agricultural Ecosystems	✓	✓	✓	✓	data management & governance, interoperability, data privacy and security, knowledge gaps, and infrastructure limitations	✓	✓	✓

Per., Perception Layer; Net., Network layer; Ed., Edge Layer; RL, Reaction Layer; CLC, Cross-Layer Coordination; FA, Formal Algorithm.

2.1 IoT-centric security approaches

The largest body of work addresses security at the device and communication level. Within this stream three concerns appear repeatedly. The first is establishing device identity and protecting transmission: Alyahya et al. propose tamper-resistant authentication schemes for agricultural IoT devices, and Vangala et al. address security-enhanced communication protocols at the network layer [16, 67]. The second focuses on detecting malicious activity in deployed IoT devices. This is addressed through intrusion detection for agricultural IoT environments [36] and threat intelligence methods for agricultural infrastructures [22]. The third is architectural: Gupta et al. and Demestichas et al. survey IoT security mechanisms and smart farming architectures. They identify integration and cyber risk as open concerns [30, 41]. Agarwal et al. provide an empirical study of Cybersecurity in a smart dairy farming system. They build a testbed using real agricultural hardware and identify several security weaknesses. These include cleartext credential transmission over untrusted web interfaces, passwordless root access to an Internet-connected ventilation controller, weak API (Application Programming Interfaces) authorisation that exposes data across tenants, and cloneable RFID tags controlling the automated segregation gate [7].

Two findings are especially relevant to this work. First, the farm workstation is a single point of failure because all control logic runs on the workstation rather than on the controllers. If the workstation fails, the automated gates stop working. Second, the system uses a flat network with no firewalls or DMZs. It also combines different communication protocols, including Ethernet, CAN bus, Modbus, Wi-Fi, Bluetooth, and RFID. This shows that the interoperability problem (will be described in CH3) is a practical security issue. The study also shows that many recommended security controls are missing in real agricultural systems. In addition, an attack at one layer can affect other layers. However, each vulnerability is analysed separately, and the study does not propose a coordinated response mechanism.

What unites these contributions is also what bounds them. All operate within IoT devices and communication level, as Table 1 shows, and all treat detection as a terminal output rather than as an input to a response process. A device that fails authentication is flagged, blocked, or reported; none of the reviewed approaches specifies what the Edge or Cloud should do in consequence. The result is that device-level security is well developed while the transition from a detected event to a coordinated countermeasure is left unspecified. It is precisely the transition that matters when an attack at the device level is a first step to activity at other sections of a smart farm.

2.2 Governance and trust-oriented frameworks

Another research stream emphasizes data governance, trust, and data-sharing practices within ADEs. These studies examine legal frameworks, ethical considerations, data ownership, data sovereignty, consent mechanisms, and stakeholder trust as critical factors. They influence data sharing and the adoption of digital farming solutions. Recent conceptual frameworks have begun to integrate governance principles with trustworthiness and Cybersecurity considerations to provide structured guidance for ADEs. However, despite their strong focus on regulatory compliance, transparency, and stakeholder trust, governance-oriented approaches predominantly operate at a conceptual or policy level. They provide limited guidance on how governance rules and trust mechanisms can be technically enforced, continuously monitored, or dynamically adapted in response to cyber incidents occurring across heterogeneous infrastructure layers. As a result, the operational linkage between data governance policies and real-time Cybersecurity mechanisms remains weak, particularly in complex, multi-layer ADEs [5, 11, 21, 23, 31]. This limitation is also seen in the IoT-focused studies, but from a different direction. Governance studies explain what should be done, but they do not provide a mechanism to carry it out. In contrast, device-level studies implement security controls, but they do not include a policy layer to guide and coordinate these controls.

2.3 Cloud and edge security in smart farming

With the increasing reliance on cloud- and edge-based infrastructures in smart farming, several studies address security and privacy challenges in data storage, processing, and access control at these levels. Existing approaches primarily focus on securing cloud platforms through standardized controls, access management, and compliance mechanisms, as well as applying edge computing for local anomaly detection and data filtering [4, 8]. However, these solutions typically operate at isolated infrastructure levels and do not provide mechanisms for coordinated security enforcement across devices, network, edge, and cloud levels. In particular, cloud- and edge-centric approaches rarely specify how security events detected at one layer should trigger adaptive responses across the broader ADE. This lack of cross-layer operational coordination limits the effectiveness of cloud and edge security mechanisms in dynamic and heterogeneous agricultural environments [4, 8, 10]. These works strengthen infrastructure-level security at edge and cloud/application layers but do not define coordinated cross-layer enforcement mechanisms. As with the preceding streams, the missing element is not a control but a decision procedure that determines which controls to invoke and where.

2.4 Ecosystem-level cybersecurity approaches

A smaller stream widens the unit of analysis from the device to the ecosystem. Duncan et al. introduce cyberbiosecurity, an interdisciplinary framing at the intersection of Cybersecurity, biosecurity, and food system protection, which expands the threat landscape across production systems, supply chains, and bioeconomic infrastructures but remains policy-oriented and defines no mechanism for adaptive response [32]. Aliyu and Liu narrow the scope to a specific mechanism, using decentralised ledgers and smart contracts to protect data integrity against poisoning and tampering, which secures transactions but leaves the real-time management of perception, edge, and cloud components outside its remit [15]. QUILLAQUA integrates sensing, artificial intelligence, Cybersecurity, and sustainability into a functional model for hydroponic production, achieving the broadest infrastructural coverage of the reviewed systems [24]. Maraveas et al. survey threats and countermeasures across the perception, network, edge, and cloud layers, identifying artificial intelligence, blockchain, quantum computing, awareness training, and regulation as mitigation directions [50]. Kaviya et al. study the agricultural ecosystem from the farmer's perspective rather than the infrastructure perspective. They review 78 studies and group the main threats into ransomware, phishing and social engineering, device vulnerabilities, denial-of-service attacks, data breaches, supply-chain risks, and insider threats. They also summarise national policy frameworks and practical security recommendations for farmers [45].

Unlike previous surveys, their study focuses on the human side of Cybersecurity. It highlights problems such as farmers leaving digital marketplaces after cyber incidents, long recovery times for small farms, and the importance of farmers' digital skills for using security controls correctly. These findings support the identified challenges in this study (CH5 and CH6) as important security challenges. However, the proposed measures are recommendations for individual users. The study does not provide a mechanism to enforce them. It also does not explain how the measures should be triggered, coordinated, or updated during an ongoing cyberattack.

The studies in Table 1 show that covering more layers does not automatically result in cross-layer coordination. QUILLAQUA spans all four infrastructure layers yet introduces no orchestration mechanism and no formalised decision procedure. The survey of Maraveas et al. covers all four layers of ADE while cataloguing mitigation measures rather than specifying how they are sequenced or triggered. Kaviya et al. extend the analysis to the human and policy dimensions without introducing a technical mechanism at all. Extending an architecture across layers, or across stakeholder categories, therefore does not by itself produce a coordinated response; that requires a component whose function is to decide, and none of the reviewed approaches designates one.

2.5 Limitations of existing approaches and research gap

Despite the significant contributions made in the IoT-, governance-, cloud/edge-, and ecosystem-based research approaches, the existing research in the field of Agriculture 4.0 Cybersecurity is structurally fragmented in nature. The IoT-based research contributes significantly to the security of IoT devices. The governance-oriented research focuses on policy, compliance, and trust issues. And the cloud/edge-centric research provides robust security for the cloud and edge environments. Most governance-oriented models remain largely conceptual and offer limited mechanisms for technical implementation. They also lack continuous monitoring or adaptive policy execution. Likewise, infrastructure-focused approaches often fail to clearly explain how security events are detected at one layer. They also are unable to trigger coordinated responses across other layers. The link between governance policies, security controls, and real-time incident response is still weak in complex agricultural settings. This lack of interconnection makes it hard to tackle key challenges. These goals include enforcing governance, improving interoperability, and meeting infrastructure needs [5, 8, 11, 41, 62, 67].

Consequently, existing research has not yet produced a single framework that can coordinate prevention, detection, and mitigation across ADE's layers adaptively. Without a clear reaction or orchestration mechanism, it is difficult to convert security requirements and governance guidelines into timely and effective actions [5, 8, 11, 41, 62, 67]. This unresolved gap highlights the need for a multi-layer cybersecurity framework that explicitly formalizes reaction as a distinct architectural layer, enabling coordinated, policy-aware, and cross-domain mitigation across heterogeneous ADE environments, as proposed in this study. Table 1 summarizes the reviewed studies according to their layer-level coverage within the ADE architecture.

2.6 Advancement over the state of the art

The review presented in Sects. 2.1–2.4 reveals a consistent structural gap across all existing approaches to Cybersecurity in Agriculture 4.0. As summarised in Table 1, none of the Agriculture 4.0 approaches reviewed here simultaneously satisfies three conditions that are essential for effective Cybersecurity in heterogeneous ADE settings: the presence of a cross-layer orchestration mechanism, the provision of cross-layer coordination that links security events and responses across all four layers, and the formalisation of the coordination logic through a decision algorithm that can be systematically evaluated and extended.

IoT-centric approaches [16, 30, 36, 41, 67] provide robust device- and network-level security but are limited to the

Perception and Network layers. They do not define how a threat detected at the device level should trigger coordinated responses at the next layers. Governance-oriented frameworks [5, 11, 21, 23, 31] address regulatory compliance and stakeholder trust but remain at a conceptual level. They offer no technical mechanism for enforcing governance policies in real time during a cyber incident. Cloud and edge security approaches [4, 8, 10] strengthen infrastructure-level controls but treat each layer in isolation, without specifying how a threat contained at the Edge layer should update access policies at the Cloud layer or alert devices at the Perception layer. Even the most integrated ecosystem-level approaches, such as QUILLAQUA [24], which cover all four infrastructure layers, do not introduce a dedicated orchestration mechanism and do not provide a formalised method governing cross-layer decision-making.

The framework proposed in this study addresses all three gaps simultaneously. The *Reaction Layer* is introduced as a dedicated fifth architectural component that sits alongside the other layers and acts as a central coordination engine, transitioning the system between Prevention, Detection, Mitigation, and Audit states in response to detected threats. This orchestration is what existing approaches lack: a single component that translates governance policies, security events, and threat classifications into concrete, layer-specific countermeasures in real time. The innovation of this study therefore lies not in securing any individual layer in isolation, but in the design of a coordination architecture that makes the security of each layer contingent on the security state of all others. The coordination logic is further formalised through the *Reaction Layer* Cross-Layer Coordination (RLCC) algorithm of Sect. 8.3, which specifies the response process in pseudocode and links each mitigation action to one or more of the challenges (CH1–CH6). This combination—a novel architectural component, a cross-layer coordination model, and a formal decision algorithm—constitutes the primary advancement of this study over the existing state of the art in agricultural Cybersecurity.

3 Methodology for challenge identification and framework development

This study uses a qualitative, literature-based methodology to identify the main Cybersecurity challenges and to develop the proposed multi-layer framework for ADEs. The methodology is based on a structured narrative and thematic review of existing studies. This approach makes it possible to combine technical, governance, and human perspectives.

A narrative approach was chosen instead of a systematic review for two reasons: First, the goal of this study is to develop a conceptual security framework, not to collect all available evidence for a specific research question. Second,

many important sources, such as data protection regulations, agricultural data-sharing codes of conduct, security standards, and institutional reports, are not included in academic databases. These sources were excluded by a strict systematic review. Therefore, a narrative approach was deemed more suitable for this study.

The methodology includes three stages: (i) challenge identification, (ii) challenge prioritization, and (iii) mapping the challenges to the ADE layers.

Stage 1: Challenge identification. Candidate challenges were identified by analysing common themes in four research areas: “Cybersecurity in Agriculture 4.0”, “agricultural data governance”, “Internet of Things security”, and “cloud and edge computing in smart farming”. In addition to research papers, regulations, data governance guidelines, and agricultural data-sharing codes of conduct were also reviewed. These documents helped identify governance and compliance issues. The identified issues were grouped into common themes. Three selection principles were used to choose the final challenges. First, the issue had to appear in more than one source and more than one application domain. Second, it had to be relevant to the ADE architecture, its operation, or its human aspects. Third, it had to be possible to address the issue through architectural design. Issues that satisfied all three principles were grouped into six main challenges (CH1–CH6). These challenges cover the technical, governance, and human aspects of cybersecurity in agricultural data sharing.

Stage 2: Challenge prioritization. The identified challenges were ranked according to their effect on the security, trustworthiness, and reliability of ADEs. Three factors were considered: the impact of the challenge, the level of risk, and its effect across multiple layers. Challenges with greater impact, higher risk, and wider cross-layer effects received higher priority.

Stage 3: Mapping challenges to ADE layers. Each challenge was mapped to the ADE layers using three criteria. The first criterion was the layer where the challenge appears. The second was the layer where the challenge can be controlled or reduced. The third was whether the challenge can affect other layers. Challenges that influence more than one layer were mapped to multiple layers. The results show that Cybersecurity challenges in Agriculture 4.0 are not limited to a single layer. Instead, they affect several layers and require coordinated protection. This finding supports the introduction of the proposed *Reaction Layer*. The *Reaction Layer* coordinates security actions across all ADE layers. Table 2 summarises the identified challenges, their priority, layer mapping, and cyber attacks related to each challenge.

Methodological scope and limitations. This study is a structured narrative review rather than a systematic literature review. It does not use predefined search strings, screening counts, or a PRISMA protocol. Therefore, it does not claim

Table 2 Methodology for identifying, prioritizing, mapping challenges to ADE layers, and associated cyber threats

ID	Challenge focus	Impact scope	Priority rationale	Mapped ADE layer(s)	Cyber attacks driven by the challenge
CH1	Data management strategy and accountability	Ecosystem-wide	Lack of clarity affects data usage, responsibility, and stakeholder trust	Cloud/Application, Edge	Data tampering, data poisoning, insider manipulation, metadata corruption
CH2	Data governance and regulatory oversight	Ecosystem-wide	Defines rules, roles, and accountability across stakeholders	Cloud/Application	Privilege escalation, identity spoofing, access control bypass, policy misconfiguration
CH3	Integration and interoperability of technologies	Cross-layer	Poor integration leads to cascading failures and data inconsistency	Cloud/Application, Edge, Network, Perception	API exploitation, Man-in-the-Middle (MITM), protocol attacks, replay attacks, supply chain attacks
CH4	Data security, privacy, and cyber threats	Ecosystem-wide	High risk to data confidentiality, integrity, authentication, availability, and non-repudiation	Perception, Network, Edge, Cloud/Application	Ransomware, DDoS (Distributed Denial-of-Service), data exfiltration, IoT malware, authentication attacks
CH5	Knowledge sharing and inclusiveness	Human-centric	Low awareness reduces adoption and weakens security practices	Cloud/Application	Phishing, social engineering, credential harvesting, configuration errors
CH6	Equipment and facilities constraints	Infrastructure-level	Resource limitations affect availability and system resilience	Perception, Network, Edge, Cloud/Application	Denial-of-Service (DoS), firmware exploitation, physical tampering, IoT botnet recruitment

to include every published study. The risk of subjectivity was reduced in two ways. First, only issues reported in more than one source and more than one application domain were included. Second, the six identified challenges (CH1–CH6) are consistent with the findings of the survey papers discussed in Sect. 2. A fully systematic mapping study can be considered as future work. The six identified challenges are discussed in detail in Sect. 4. The next section explains each challenge, its causes, its practical effects, and its importance in the ADE.

Section 8.5 additionally presents a targeted comparison against orchestration approaches from adjacent domains. That comparison is purposive and bounded in scope, serving to position the contribution of this study rather than to sur-

vey the orchestration literature exhaustively; it is therefore consistent with the narrative methodology adopted here.

4 Structural challenges driving cybersecurity risks in agriculture 4.0

Building stakeholder trust and awareness of innovative Agri-food 4.0 technology is essential for encouraging acceptance, practical implementation, and preventing resistance and skepticism that may impede progress in the sector. However, to gain common acceptance, several challenges must be overcome, which are identified through the methodology

described in the previous section and discussed in this section [51].

4.1 CH1–Straightforward data management strategy

The lack of a clearly defined strategy for data utilization creates major obstacles in realizing its full value [52]. On 23 April 2018 a joint EU (European Union) Code of Conduct on agricultural data sharing was launched by a coalition of associations from the EU agri-food chain called the EU Code of Conduct on Agricultural Data Sharing by Contractual Agreement. While not mandatory, the guidelines and regulations urge agribusinesses to provide transparency on several aspects of using farmer data to improve farmers' ability to make informed decisions and assuring them to share their data with tech service providers, supply chains, and other stakeholders in the ecosystem [23]. However, despite these initiatives, uncertainty remains among farmers regarding the effectiveness of existing regulations and practices in safeguarding their data, along with concerns about potential vulnerabilities and data breaches [19].

To increase the criticality of this situation, the interconnection and interaction of agricultural features expose them to cybersecurity risks if proper countermeasures cannot be implemented [64].

In this context, recent European Union agricultural policies, including the Data Governance Act, seek to clarify this activity to strengthen trust in data intermediaries and to enhance data sharing across the EU. However, existing discussions often overlook how data may shift between personal, non-personal, or sensitive categories across different stages of the ecosystem and rarely clarify who is accountable for securing data before, during, and after a cyber incident. Questions also persist regarding whether liability may fall on farmers, service providers, technology vendors, or other intermediaries, and what legal or procedural steps must be followed after a breach [28]. Variations in standards across countries, sectors, regions, and organizations have created a fragmented landscape of policies, frameworks, and operational practices that pose certain challenges to the evolution of a common risk management framework [52].

4.2 CH2–Data governance

When it comes to sharing data from farms and agribusinesses that are willing to engage in a more open data space [23], having good practices for managing data can be even more important [31]. However, within Agriculture 4.0, data governance must extend beyond operational data handling and include strategic oversight, long-term planning, and

the establishment of rules that define how information is accessed, protected, and ethically used [11]. It should cover a broad set of elements, including policies, standards, ethical issues, and risk management practices applicable to both institutions and individuals, particularly in relation to sensitive data and data sharing arrangements. Devare et al. [31] and Rahbar Nodehi et al. [62] emphasize the need to create a trustworthy data environment, enabling users to access and utilize data confidently [68].

While data management focuses on how data is stored, structured, and moved [11], data governance functions as a higher-level decision framework that specifies responsibilities, permissions, and actions over the entire data lifecycle. This activity differs from data management and operates alongside data management, rather than replacing it. These governance mechanisms are typically implemented through predefined guidelines. These guidelines establish who can access data, under what conditions, and according to what procedures [11, 62].

4.3 CH3–Integration and interoperability

The agricultural sector is undergoing a major transformation. Climate change, population growth, and new technologies are making agricultural ecosystems more complex. Technologies such as blockchain, Artificial Intelligence (AI), and the IoT offer many opportunities. However, integrating these technologies remains a major challenge. Practical deployment raises issues such as data reliability and increased exposure of agricultural systems to cyberattacks.

Another challenge is the lack of common governance structures and widely accepted integration standards. Without universal standards, interoperability depends on middleware that hides differences between devices, communication protocols, and networks. For example, agricultural systems may translate Message Queuing Telemetry Transport (MQTT), and Constrained Application Protocol (CoAP) traffic at gateways into a common context representation [53]. As a result, each gateway is both an integration point and a trust boundary. This makes CH3 a security challenge as well as an engineering challenge.

Differences in data formats and system designs make integration even more difficult. These differences reduce trust in data sharing and interoperability. They can also limit cooperation within the Agricultural Ecosystem (AE), especially when organisations share critical resources or develop joint initiatives. Therefore, harmonised data models and interoperability standards are essential for integrating different technologies and improving their effectiveness across the ecosystem [26, 51].

4.4 CH4–Data security, privacy, and cyber threat impacts

Data security and privacy are among the critical determinants of trust in data-driven agricultural environments and play an important role in farmers' willingness to cooperate with data-sharing schemes. Past research suggests that farmers' perceptions and willingness to cooperate with data-sharing schemes depend on the type of data-sharing platform's governance structure and their level of perceived trust in the entities involved in the process [19]. In response to these concerns, the European Union has emphasized the role of trust and transparency in enabling effective data-sharing processes in ADEs, recognizing farmers as critical actors in the governance of ADEs [21].

In addition to data confidentiality and ownership, Agriculture 4.0 is susceptible to a variety of cybersecurity threats that affect not only data security but also the availability, integrity, and reliability of agricultural operations. Past research suggests that cyberattacks on Agriculture 4.0 can result in service disruptions, loss of operational awareness, manipulation of control processes, and other unforeseen consequences on Agriculture 4.0 operations [36, 41]. These cybersecurity threats target different levels of ADEs, including perception-layer devices, network communications, and cloud-based management platforms [8].

4.5 CH5–Knowledge sharing and inclusiveness

Many farmers and other stakeholders have limited knowledge of digital agricultural technologies. This makes it difficult for them to use ADEs safely and effectively. When users do not understand how data are collected, transmitted, and processed, they are less likely to trust or adopt these systems. This problem is especially common among small farms and users with limited experience with digital technologies. This challenge can be reduced through training, advisory services, demonstration farms, and easy-to-understand technical documentation. Collaborative platforms and farmer networks also help users to share knowledge and practical experiences. These activities reduce the digital divide and support the wider adoption of secure agricultural technologies [49].

Limited cybersecurity knowledge also creates security risks. Users who do not know how to manage passwords, configure devices, or recognize phishing messages are more likely to become victims of cyberattacks. Incorrect system configuration can also create security weaknesses. Therefore, knowledge sharing and inclusiveness are not only important for technology adoption. They are also essential for cybersecurity because technical protection is effective only when users know how to use it correctly [50].

4.6 CH6–Equipment and facilities constraints

Digital agriculture requires sensors, communication networks, software platforms, and other equipment. The cost of purchasing and deploying these technologies can be too high for many farms, especially in rural areas. In addition, maintenance, security updates, and system operations create ongoing costs that may not be affordable for small agricultural enterprises. Rural areas also face infrastructure problems, such as unstable Internet connections and unreliable power supplies. These conditions make it difficult to operate connected systems continuously. Cost-effective technologies, shared infrastructure, government support, and mobile-based solutions can help reduce these challenges, especially where network connectivity is limited [3].

These limitations also affect cybersecurity. Limited resources often delay software updates and patches. Older devices and unsupported firmware remain in use for longer periods, thereby increasing security risks. Limited monitoring also makes attacks harder to detect. In addition, many low-cost devices cannot support advanced encryption or continuous security monitoring because of their limited computing resources. As a result, they are more vulnerable to denial-of-service attacks, firmware attacks, physical tampering, and IoT botnets. Therefore, equipment and infrastructure constraints affect both the adoption of ADEs and the level of cybersecurity that these systems can provide [30, 50].

4.7 Comparison of existing approaches with respect to identified challenges

Section 2 examined the existing works from an architectural perspective. It considered which ADE layers are covered by each research stream and whether the approaches provide an orchestration mechanism, cross-layer coordination, and a formal decision-making process (Table 1). This subsection examines the same research from a different perspective: how well each research stream addresses the challenges CH1–CH6 identified in this section. These two views complement each other. The first shows “where existing approaches operate in the architecture”, while the second shows “which challenges remain unresolved” and “why these approaches cannot work together as a coordinated defence”.

Research on IoT security mainly addresses the technical aspects of the ecosystem. It protects devices and communication channels using authentication and encryption and detects intrusion attempts at the device and network levels. This research provides strong support for data security, privacy, and cyber threats (CH4). However, it provides limited support for interoperability and system integration (CH3), human-related factors such as knowledge sharing and inclusiveness (CH5), and infrastructure limitations (CH6). As a result, these security measures mainly protect individual

devices rather than the whole ecosystem in which they operate.

Governance- and trust-based frameworks mainly address the strategic and regulatory aspects of agricultural data sharing. Therefore, they partly address data management strategy (CH1) and data governance (CH2). These models emphasise trust, transparency, and compliance. However, they mainly remain at a conceptual level and do not provide a clear mechanism for implementing or monitoring governance rules during a cyber incident. Therefore, operational security threats (CH4) and coordination between different layers are not sufficiently addressed.

Cloud- and edge-based approaches mainly focus on infrastructure-level security controls, such as access management, anomaly detection, and secure data processing. They support CH3 at the platform level and address CH4 within the layer they protect. However, they do not explain “how a response started in one layer can be transferred to the perception, network, edge, and cloud layers as one coordinated action”. They also do not consider stakeholder awareness (CH5) or infrastructure limitations in rural areas (CH6).

Ecosystem-level approaches provide the broadest coverage of the identified challenges. QUILLAQUA [24] addresses integration, security, and governance issues (CH2–CH4). The reviews by Maraveas et al. [50] and Kaviya et al. [45] also consider knowledge gaps (CH5) and infrastructure limitations (CH6). However, their contribution is mainly descriptive. They list threats and possible countermeasures for each challenge, but they do not provide a mechanism to decide “which countermeasure should be used”, “at which layer”, or “in what order after a threat is detected”.

These limitations are not only related to the coverage of individual challenges. Even if these four research streams were combined in one ADE, they would not automatically provide a coordinated response. This is because each stream produces an output that the other streams cannot directly use. IoT-based approaches can produce an alert about a compromised device, but they do not specify which component should receive or use this information. Governance approaches can provide a rule that defines an obligation, but they do not identify a runtime component that can enforce the rule during a security incident. Cloud- and edge-based approaches can produce a containment action, but this action usually remains within the layer where it is triggered and does not provide instructions for the other layers. These research streams also work at different timescales. Governance rules are usually developed, negotiated, and reviewed over longer contractual periods, while threat detection and containment operate at the timescale of network traffic. None of the reviewed approaches provides a clear mechanism for connecting these two timescales. Therefore, the main problem is not a lack of coverage, but a lack of coordination and composition between the existing approaches. This identifies

the missing element: a component that can receive detection results, evaluate them against the active policies, and provide mitigation instructions for the appropriate layers. This gap motivates the proposed framework, which introduces a dedicated *Reaction Layer* to coordinate prevention, detection, and mitigation across the ADE. The proposed layer explicitly considers technical, governance, and human-centric challenges. The remainder of the paper provides a detailed analysis of the identified challenges and the mechanisms proposed to address them.

5 The shift from data management toward data governance

Data has increasingly become a valuable and strategic resource worldwide [52]. In today’s agricultural environment, realizing its full potential requires a coordinated system in which stakeholders across the AE work together and share information. It generates insights and economic value [68]. Data governance is essential for ensuring that ADE assets are accessible, shareable, secure, reliable, and protected from misuse [31]. A data ecosystem (DE), such as a complicated network, consists of multiple actors participating in various data-related activities, such as searching, storing, sharing, accessing, and recycling data. This collaboration promotes innovation, generates value, and supports new business initiatives [68].

Although data governance has traditionally been informal, it remains relatively understudied in academic research and limited in practical industry adoption [11].

Although data governance has traditionally been informal, it remains relatively understudied in academic research and limited in practical industry adoption [11]. The United Nations Conference on Trade and Development (UNCTAD) has noted that there are various frameworks and strategies for governing data and regulating data flows, with minimal consensus at both the global and regional levels [52]. The need for a unified data governance structure arises from a complex environment where policy and implementation intersect, addressing areas such as data management, ethics, and protection [52].

5.1 Defining data governance

Researchers have offered various definitions of data governance.

- The Data Governance Institute (DGI) defines it as *a system of decision rights and responsibilities for information-related processes, implemented based on agreed models that specify who can perform which*

actions with what data, when, under what conditions, and using which methods [11].

- Gwen Thomas of the Data Governance Institute defines data governance simply as “making decisions and exerting authority over data issues” [40].
- Tony Fisher, in his book *The Data Asset*, describes data governance as “a methodology and philosophy for benefiting from your data. It is not a program or technology that will fix a problem. It is a mindset about establishing a culture where quality is achieved, maintained, valued, and used to drive the business” [40].
- The Data Management Association (DAMA) defines data governance as “the act of exercising authority, control, and collaborative decision-making (planning, monitoring, and enforcement) over managing data assets” [11].

In a systematic literature review, Abraham et al. concluded that no single, universally accepted definition of data governance exists in the academic or professional literature. Therefore, they identified common features using open coding. Their analysis led to the following definition: *Data governance specifies a cross-functional framework for managing data as a strategic enterprise asset, determining decision rights and responsibilities for an organization’s data-related decisions. It also formalizes data policies, standards, procedures, and monitors compliance* [5].

According to this definition, as shown in Fig. 1, data governance provides the structure and formalization required to manage data. The proposed framework includes:

- Antecedents: Conditions that shape the adoption and implementation of data governance, such as organizational structure, regulatory requirements, technological readiness, leadership support, and organizational culture toward data management.
- Consequences: The outcomes of implementing data governance [5].

Organizations must clearly determine the scope of data governance. At the core of data governance, organizations must define what needs to be governed (i.e., the scope of data), who governs the data (i.e., roles and governance bodies), and what decisions must be made in data-related areas (e.g., data quality, security, architecture, lifecycle, metadata, and infrastructure) [6, 14].

5.2 Data governance in ADEs

In the context of ADEs, data governance refers to the establishment of principles and rules that all parties must follow to create a trustworthy data environment, enabling users to

access and utilize data confidently [68]. The key principles of data governance in ADEs and data spaces include the following:

- Defining terms for data participation and sharing, including confidentiality, monetization, and property rights.
- Protecting data owners’ rights, particularly through consent mechanisms for accessing and using data.
- Ensuring adherence to security regulations to prevent unauthorized access and misuse, especially for sensitive data.
- Implementing specific measures for particular data types, such as data anonymization, while preserving data quality.

5.3 Global initiatives and challenges

Several global initiatives aim to establish trust in ADE. For instance, on April 23, 2018, the EU Code of Conduct on Agricultural Data Sharing by Contractual Agreement was launched by a coalition of EU agri-food associations [23]. In February 2020, the European Commission introduced a new data strategy, proposing a European data space that includes sector-specific data spaces, such as the agricultural sector. The agricultural data space aims to support the creation of a smart, innovative, and sustainable agri-food system [47]. However, these initiatives have not fully addressed key issues such as trust in data sharing, accessibility, and data portability. Consequently, a comprehensive and standardized framework for data governance in the agricultural sector remains absent [68].

6 The agricultural data-sharing ecosystem architecture

The ADE has five layers, as shown in Fig. 2. These layers include: (a) the perception (Physical) layer; (b) the network layer; (c) the edge/fog (Processing) layer; (d) the cloud-based applications or services layer; and (e) the *Reaction Layer*.

The perception layer includes sensors, GPS (Global Positioning System), RFID (Radio-Frequency Identification) tags, cameras, actuators, gateways, and other devices distributed across agricultural farms are responsible for collecting and modifying data from the farm environment. These devices lack the computational capacity to process or store data; therefore, they connect to edge or cloud resources through the network layer for data processing [18]. The edge/fog layer is close to end-users and farms for local real-time processing. This approach decreases the computational load on centralized cloud resources and network traffic. It also comprises several edge nodes, each signifying a gateway

Fig. 1 The conceptual framework for Data Governance [5]

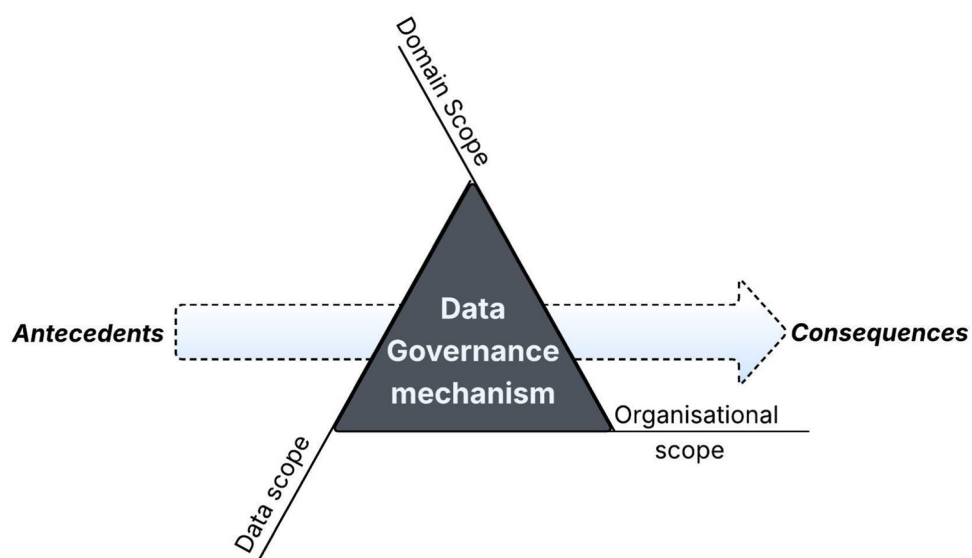
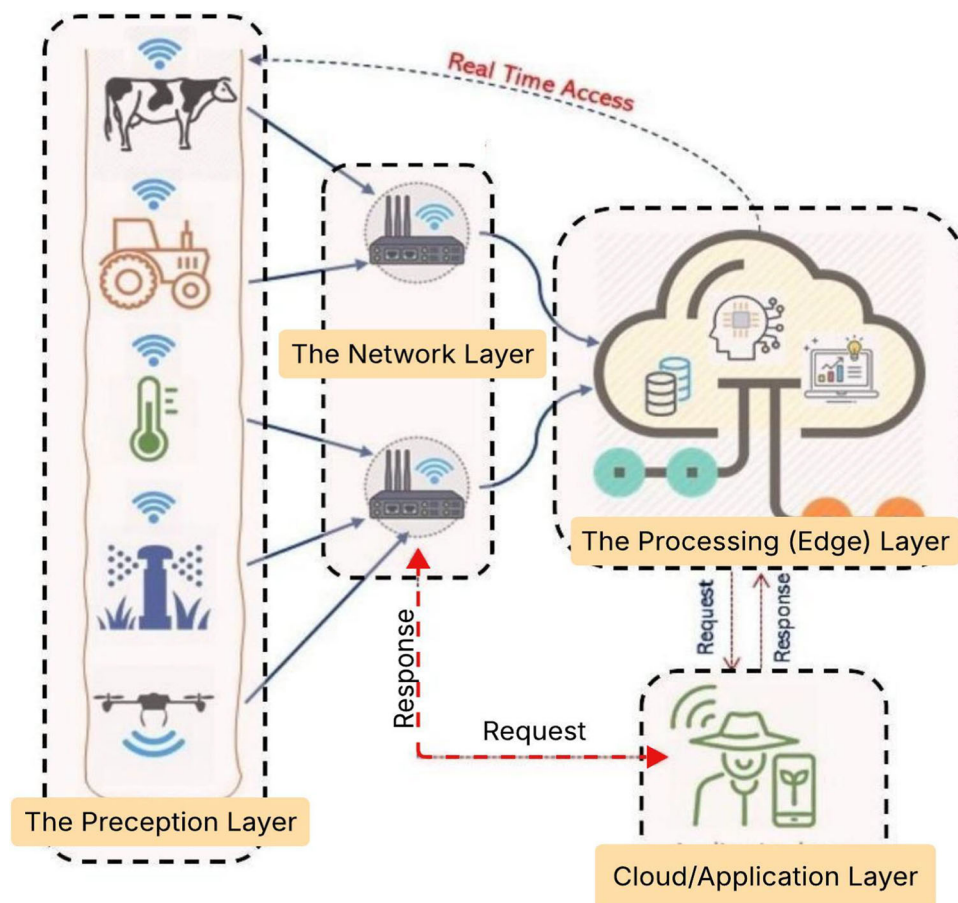


Fig. 2 The layered structure of Agricultural Data-sharing Ecosystem



that supports functions such as data acquisition, aggregation, filtering, encryption, and real-time stream processing; security monitoring and detection, forecasting, and real-time decision assistance [41].

In cybersecurity, the fundamental elements of protecting networks against cyber-attacks consist of Prevention, Detection, and Mitigation components [25]. Since cybersecurity directly impacts all layers of this ecosystem, appropriate responses must be implemented in real-time in the event of a security incident. These responses vary depending on the specific layers affected. To clarify this concept, by default, the ecosystem operates in a “Prevention” mode. This means that regulatory frameworks such as GDPR (General Data Protection Regulation), FAIR (Findable, Accessible, Interoperable, Reusable) principles, and the EU Code of Conduct impose specific security requirements on agricultural data-sharing systems [65], including authentication, authorization, data integrity, data reliability, availability, non-repudiation, trust, and privacy, which must be followed by agricultural data-sharing systems [61]. Failure to meet these requirements could impede the widespread adoption of these advancements [66].

The most common security requirements in distributed networks that are often targeted by attackers are Confidentiality, Authentication, Data Integrity, Availability, Non-repudiation, and Privacy [30, 61]. Table 3 shows which of the security requirements as goals are violated by each cyber risk in the AEs [67].

However, if a cyber-attack is detected by the Processing Layer, the system transitions from the default “Prevention” mode to a “Mitigation” mode. At this stage, the *Reaction Layer* is activated and integrated into the ecosystem to deploy appropriate countermeasures aimed at reducing the impact of the attack. The primary goal of this phase is to contain the threat, minimize damage, and restore system stability as efficiently as possible.

7 Inter-connectivity and security issues in agriculture 4.0

While the advancement of modern technologies in Agriculture 4.0 offers new opportunities for improving agricultural output [58], a plethora of new Cybersecurity risks have emerged due to the installation of many IoT devices in open fields [66] to monitor and share on-farm data in real-time [65]. Numerous new technologies in the agricultural sector are not originally designed for this industry but rather adapted from existing ones, resulting in a lack of focus on security concerns [64], such as IoT devices, smart vehicles, drones, edge cloud, and wireless communication [41] integrated into today’s farming to perform specific agricultural tasks across an ecosystem. These complex systems are often developed

Table 3 Security functionality, cyber attacks, and impacts in Agriculture 4.0 [67]

Goal	Impacts	Agricultural cybersecurity risks
Confidentiality	Location tracking; Eavesdropping; Data theft/breach; Traffic Analysis; Man-in-the-middle (MITM); Known-key attack	Unauthorized access to sensitive Agri-data can result in loss of proprietary information, compromised competitive advantage, and privacy violations for farmers
Integrity	Data pollution/poisoning; Data falsification; Data injection; Ephemeral Secret Leakage (ESL); MIM	Altered or manipulated Agri-data can lead to incorrect decision-making, impacting crop yields, resource allocation, and farm operations
Authentication	Sybil; Impersonation; MITM; Replay; Session Hijacking	Unauthorized entities may access agricultural systems, leading to Agri-data manipulation, service disruptions, and potential harm to crops, livestock, and supply chains
Availability	Channel interference; Physical capture & damage; DoS/DDoS; Cyber Aggression	Disruptions in system availability can halt operations, causing financial losses, supply chain interruptions, and reduced stakeholder trust
Non-repudiation	Repudiation attacks; Malicious code attack	The inability to verify actions can lead to disputes, fraud, regulatory non-compliance, and Agri-data manipulation

by different vendors and produced for various environments and applications, which increases the probability of cyber-attacks [12]. Despite much research on the security of each of the existing technologies in the agricultural industry, most of the time, existing studies overlook the environmental conditions under which these technologies are used [41]. Certain Cyber risks are similar to traditional attacks on network and computer systems, and in some cases, they are specific to digital environments, which open up new vulnerabilities

and security issues [12]. Hence, security standards must be continuously updated to keep up with the rapid evolution of digital technologies [10]. Standards and regulations provide a comprehensive framework that directs stakeholders to implement security measures for different assets, such as data, devices, and applications, at various levels. Currently, no specific security framework has been created for Agriculture 4.0. The NIST Cybersecurity Framework (CSF) and the International Standards Organization's ISO 31000, as the frameworks that address IT and operational security, may serve as a useful foundation for designing a Cybersecurity framework tailored to this ecosystem [12]. Note that the Prevention, Detection, Mitigation elements' considerations together form a comprehensive security framework that must be addressed as a whole when protecting inter-connectivity from various types of attacks [25].

7.1 Legal considerations in data sharing in agricultural infrastructures

IoT devices utilized for collecting data on farms are vulnerable to various security and privacy breaches, and the data they transmit may not be trustworthy for subsequent analysis. IoT devices lack sufficient memory and power to be adequately protected by conventional cybersecurity measures. Therefore, the development of a robust and efficient security system is essential for Agriculture 4.0 [16]. Although many security management frameworks exist for various types of cyber risks, not all apply to IoT devices [17]. The common security frameworks in the field of IoT devices are listed in Table 4, which highlights how each framework applies to cybersecurity in agricultural IoT (Agri-IoT) environments [17].

These frameworks provide foundational information on IoT security best practices, including guidance on monitoring systems, responding to disaster incidents, deploying new applications, periodically evaluating implemented security, isolating systems, restricting physical and logical system access, acceptable use policies, enforcing proper security and privacy policies, and compliance [17].

When it comes to data sharing at higher levels in the ADE, ensuring the proper security of agricultural data requires standardizing all aspects of sharing to prevent unauthorized access. This raises questions about who should be responsible for ensuring data security after an intrusion is detected. Can the responsibility for security breaches be attributed to the farmer, technology and Internet provider, or other separate entities in the ADE? How will the mitigation process work? Is it necessary to consider penalties or alternative legal actions against possible data misuse [28]? Therefore, to meet these requirements, frameworks have been developed to help ADE establish the necessary security processes, procedures, and tools to evaluate, track, and reduce cyber risks,

as well as enhance the overall security posture [54]. Unfortunately, the lack of uniformity in existing security standards leaves farmers' level of risk in the hands of independent companies, often communicated through complex and lengthy documents filled with technical language [28].

7.2 Legal considerations in cloud-based agriculture

It is worth noting that these Cybersecurity assessment frameworks are too general for cloud computing. There is no clear industrial consensus on how cloud risks should be evaluated. It is mainly due to the absence of comprehensive frameworks and the rapidly evolving nature of cloud technologies. Similar to IoT, there are few quantitative methods for managing Cybersecurity risks, and none of the existing frameworks address the security of the IoT ecosystem [10].

In cloud settings, a security concern refers to any adverse event that may compromise digital assets hosted within the cloud infrastructure. The assets may include data, software, infrastructure, customer trust, and organizational reputation. Therefore, security concerns are categorized into four groups: a. data security, b. network and services security, c. applications security, and d. Human-related security factors. This classification is done with consideration of the most recent patterns of attacks on cloud computing systems [8]. The widespread application of cloud computing has led to the emergence of numerous cloud standards. The most common ones and their applications in Agri-cloud computing and Agri-data security are depicted in Table 5.

8 The proposed conceptual framework

The proposed Cybersecurity framework is derived from a literature-informed analysis of six key challenges (CH1–CH6) identified across the ADE. Rather than introducing an ad hoc or component-specific solution, the framework integrates governance, technical, and operational requirements into a unified and coordinated architecture. Central to this architecture is the *Reaction Layer*, which enables adaptive, cross-layer security coordination across the ADE. This section introduces the architectural foundations of the proposed framework, establishes its underlying threat assumptions, and subsequently maps the identified challenges to the ADE layers.

The multi-layer ADE architecture considered in this study explicitly acknowledges the presence of heterogeneous cybersecurity threats and vulnerabilities spanning perception, network, edge, and cloud/application layers. Addressing such threats requires a holistic and integrated cyber risk management approach that extends beyond isolated security controls. Accordingly, the proposed framework builds upon and adapts existing security practices by incorporat-

Table 4 IoT security management frameworks [17]

Title of framework	Functionality	Agricultural relevance
Cloud Security Alliance IoT Security Controls Framework (2019) Providers Cloud Security Alliance (CSA)	It includes best practices for security in connected devices, cloud environments, supporting systems, and associated data	Ensures robust cloud-based farm management and secure communications of agricultural data from cyber threats
IoT Security Maturity Model (SMM): Practitioner's Guide for Assessing and Advancing IoT Security (2019) Industrial Internet Consortium (IIC)	It introduces a secure development lifecycle (SDL) for building, deploying, and operating secure IoT solutions	Helps design secure smart irrigation and precision farming solutions by enforcing security best practices at every stage
NIST Considerations for Managing IoT Cybersecurity and Privacy Risks (NISTIR 8228) (2020) National Institute of Standards and Technology (NIST)	It provides IoT operational parameters and IoT Cybersecurity and privacy practices focusing on device security, data security, and safety	Ensures safe precision agriculture operations by protecting agricultural machinery and processing sensitive farm data
IoT Security Foundation (IoTSF) IoT Security Compliance Framework – IoT Security Foundation (IoTSF)	It introduces a top framework for security compliance across IoT devices and systems	Gives guidelines for deploying and maintaining secure IoT devices on farms
Standard for an Architectural Framework for the Internet of Things (IoT) (ISO/IEC 30141:2018) International Organization for Standardization (ISO)	It is a blueprint for data simplification, communication, processing, and analytics and is designed for the interoperability of complex IoT systems	Assists in structuring smart farming architectures and enabling interoperability of agricultural IoT systems
Industrial Internet Consortium (IIC) Industrial Internet Reference Architecture (IIRA)	It describes a framework for various industrial IoT domains that provides horizontal and vertical system architecture views from different layers of the system	Facilitates complex, cross-domain agricultural IoT deployments, such as integrating farm equipment, sensors, and market data networks

ing advanced access control mechanisms across all layers, prevention, detection, and mitigation of data leakage, protection against faulty or compromised sensors and IoT devices, development of industry-wide standards to foster trust among stakeholders, integration of cyber insurance strategies for financial risk mitigation, real-time visibility of connected devices, behavioural profiling of smart farming operations for anomaly detection, and the adoption of adaptive defensive strategies suitable for dynamic agricultural environments.

8.1 Threat model and cross-layer adaptation

To demonstrate how the *Reaction Layer* effectively addresses the heterogeneous threats in Agriculture 4.0, this subsection presents a tailored STRIDE (Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, Elevation of Privilege)-based threat model and explains the adaptive

mechanisms integrated within the framework. The threats in Agriculture 4.0 are very varied including IoT devices on farms, systems that combine computer parts, platforms for food supply chains and cloud-based services for analyzing data. This variety of threats means the ADE is open to kinds of attacks that can happen at different levels. This study uses the STRIDE threat model, which includes Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service and Elevation of Privilege. This model is tailored to identify risks associated with the different layers of the ADE. The STRIDE model was originally developed by Microsoft. It is widely used for looking at threats in IoT devices and systems that combine physical and computer parts because it groups common security weaknesses in a clear way. In the context of Agriculture 4.0, using STRIDE to analyze threats has been shown to be effective in finding potential threats in internet-

Table 5 Cloud security management frameworks [4]

Title of framework	Functionality	Agricultural relevance
ISO/IEC 27017 Provider Implementation (ISO)	It contains information security management tools for cloud solutions. This standard is called the “Code of Practice for Information Security controls based on ISO/IEC 27002 for cloud services”	It helps to standardize and improve the security of cloud services used in agricultural systems and farm management platforms
ISO/IEC 27018 Provider ISO	It reflects the rules for the security of cloud services and personal data protection in a public cloud	It supports the protection of farmers’ and consumers’ personal data stored and processed in agricultural cloud platforms
NIST Special Publication 800-144 Provider: NIST	It defines the rules for providing the security for federal information systems and organizations in cloud computing	It provides guidelines for secure adoption of cloud services in agricultural agencies and research institutions
Privacy in Cloud Computing Provider: NIST	It analyzes the problems of standards to be achieved with privacy requirements and cloud data privacy	It is applied to ensure privacy of agricultural data, such as farm production records and geospatial information stored in cloud environments
ITU-T X.1601 Provider: The International Telecommunication Union (ITU)	It consists of general guidelines for the security of cloud computing	It offers generic security guidance that can be tailored to cloud-based agricultural applications and services
OWASP Secure Coding Practices Provider: Open Web Application Security Project (OWASP)	It has guidelines to train developers in secure coding practices for web applications	It helps developers of agricultural web platforms to reduce software vulnerabilities that could affect farm data and services
FedRAMP Provider: U.S. Government	It recommends technological and operational procedures for standardized risk evaluation of cloud services for federal security management	It is used when agricultural agencies consume U.S. federal cloud services to ensure that security requirements are met
CSA STAR Provider: Cloud Security Alliance (CSA)	It contains documents for rating the security level of a cloud provider	It can be referenced by agricultural organizations to select cloud providers with appropriate security assurance levels for farm-management solutions and IoT deployments

connected devices, which helps to assess the security of the whole ecosystem [9].

The STRIDE-based threat model considers risks in three areas: who might attack, what needs to be protected, and how attackers might gain access. The STRIDE-based threat model

characterizes risks along three complementary dimensions: threat actors, protected assets, and potential entry points.

Threat actors: Potential adversaries include cybercriminals seeking financial gain via ransomware (e.g., dairy processing disruptions; [27]), nation-state actors targeting

supply chains for strategic disruption (cyberagrorterrorism; [41]), and insiders such as disgruntled employees exploiting legitimate access.

Assets: Critical assets within the ADE comprise sensitive agricultural data (e.g., crop yields, soil metrics, and farm management records), physical and cyber-physical devices (e.g., autonomous tractors, robotic systems, and IoT sensors at the perception layer), and digital infrastructure supporting data processing and sharing, such as edge computing nodes for real-time analytics and cloud platforms enabling large-scale data aggregation and collaboration.

Entry points and attack vectors: Cyber attacks frequently leverage vulnerabilities across multiple ADE layers. For example, spoofing attacks at the perception layer may involve impersonation of IoT sensors to inject false measurements. Such attacks can manipulate soil pH values that mislead fertilization or irrigation decisions. Tampering attacks often target network-layer communications, particularly wireless protocols deployed in remote agricultural environments. They can lead to data integrity violations. Denial-of-Service (DoS) and Distributed DoS mainly target the network and edge layers. They can disrupt the availability of live monitoring and control functions. Livestock tracking and machinery coordination, as highlighted in recent European threat landscape reports [34] are clear examples of that. At the cloud and application layers, information disclosure attacks may exploit misconfigurations or weak access control mechanisms. It can result in the leakage of proprietary or sensitive farm data. Furthermore, repudiation and elevation-of-privilege attacks can enable advanced persistent threats, such as malware infiltration of supply-chain platforms. Such attacks ultimately allow unauthorized control over autonomous agricultural machinery [4].

The proposed framework addresses these heterogeneous attack vectors through the *Reaction Layer*, which functions as a core orchestration mechanism for proactive and coordinated cross-layer responses. In smart farming settings, the *Reaction Layer* leverages CTI techniques to coordinate adjustable strategies across prevention, detection, and mitigation stages [22].

Table 6 brings these elements together in one threat–asset–mitigation matrix. Each STRIDE threat is linked to the assets it targets, the ADE layer where it appears, the security goals from Table 3 that it affects, and the RLCC sub-procedure in Algorithm 1 that responds to it. The matrix has two purposes. First, it shows the mapping between each threat and the affected security goals. This mapping is used to calculate the term $G(e)$ in the severity score presented in Sect. 8.4. Second, it makes every mitigation action traceable to a specific threat and to one or more of the challenges CH1–CH6.

- **Prevention:** Predefined security policies enforce robust controls, including multi-factor authentication (MFA)

and zero-trust mechanisms, to counter spoofing and elevation-of-privilege attacks at the perception and network layers. Cryptographic and blockchain-based data integrity mechanisms are employed to prevent tampering during supply-chain data exchange across heterogeneous platforms, thereby supporting interoperability efforts identified in CH3.

- **Detection:** Anomaly-based monitoring leverages machine learning models, such as CNN–LSTM (Convolutional Neural Network–Long Short-Term Memory) architectures for efficient DDoS detection in IoT networks [13], combined with deception technologies like honeypots to identify malicious activities and divert attackers from critical farm assets [22]. Real-time CTI feeds integrate emerging threat indicators, enabling early correlation of events such as malware campaigns or sensor-level exploits.
- **Mitigation:** Automated response mechanisms enable rapid isolation of compromised components (e.g., quarantining ransomware-infected edge devices), data rollback and recovery at the cloud layer, and activation of failover mechanisms to maintain service availability under DoS conditions. Mitigation actions are dynamically adjusted based on threat severity to minimize cascading effects; for instance, a detected spoofing attack at the perception layer can trigger immediate policy updates and access restrictions across all ADE layers.

8.1.1 Reaction layer coordination mechanism

The *Reaction Layer* works as a decision maker that follows predefined security rules. It always looks at security events, information about what’s happening, and knowledge about threats across all parts of the ADE. Instead of using separate defenses, it allows for coordinated and adaptive responses through three main jobs: looking at events, enforcing rules, and taking action across different layers.

At the perception layer, the *Reaction Layer* interfaces with sensors. It ensures the authenticity of data by monitoring device health. Additionally, it revokes the credentials of compromised devices or isolates sensors in instances of attempted spoofing or tampering. At the network layer, it enforces rules that include: a. Who can access (dynamic access control policies)? b. What filters traffic and limits it? and c. How much data can be sent (rate-limiting rules)? These help prevent denial-of-service attacks and unauthorized communication attempts. At the edge layer, it coordinates local containment actions, such as workload isolation, edge-node quarantine, and temporary service suspension. These actions prevent attacks from spreading. At the cloud and application layers, the *Reaction Layer* organizes actions like access revocation, data rollback, policy reconfiguration, and incident logging for compliance and accountability purposes.

Table 6 STRIDE threat–asset–mitigation matrix for the ADE, with traceability to the RLCC sub-procedures of Algorithm 1 (Sect. 8.3)

STRIDE (t_e)	Asset at risk	Source layer (l_e)	Goals	RLCC sub-procedure (Phase 3)	Challenges
Spoofting	IoT sensors, autonomous tractors, and robotic systems	Perception	Authentication and Integrity	Device Authentication & Sensor Isolation: Revoke Credentials, Blacklist, Isolate Sensor, Activate Backup	CH4, CH6
Tampering	Agri-data in transit; supply-chain data exchanged across heterogeneous platforms	Perception, Network	Integrity	Isolate Sensor for tampered devices; Dynamic Access Control & Traffic Filtering: BlockFlow, UpdateACL; Rollback and Validate on integrity violation	CH1, CH3, CH4
Repudiation	Records of actions on supply-chain and data-sharing platforms	Network, Cloud	Non-repudiation	Log of dropped packets and rejected API calls; StoreLog of events, actions, severity, and resolved challenges	CH2
Information disclosure	Proprietary and sensitive farm data: crop yields, soil metrics, farm management records	Network, Cloud	Confidentiality	Reject for unauthorised API calls; Access Revocation & Data Recovery: Revoke, Notify	CH2, CH4
Denial of service	Live monitoring and control functions; livestock tracking and machinery coordination; edge nodes and services	Network, Edge	Availability	BlockFlow and RateLimit; Containment & Workload Isolation: Quarantine, Migrate, Failover	CH4, CH6
Elevation of privilege	Cloud platforms and management interfaces; control over autonomous agricultural machinery	Network, Cloud	Authentication and Confidentiality	Reject and UpdateACL for unauthorised access attempts; Revoke for compromised or policy-violating tokens; policy update via Upd	CH2, CH4

This table extends the security goals of Table 3 by linking each threat to the proposed mitigation procedure. Procedure calls in Algorithm 1 are shown in a different font

The *Reaction Layer* makes decisions based on predefined security rules, real-time monitoring data, and information about CTI feeds. This allows the proposed framework to change its response plan based on how severe the threat is, which parts are affected, and what other problems it could cause. By monitoring the ADE, the *Reaction Layer* ensures that actions taken in one part are always reflected in other parts. Thus, security responses remain coordinated and consistent. The *Reaction Layer* maintains a comprehensive view of the ADE. It supports making decisions based on established rules and an understanding of potential threats.

Figure 3 provides an overview of the proposed framework and illustrates how the *Reaction Layer* orchestrates these mechanisms across the ADE layers, enabling coordinated prevention, detection, and mitigation responses to the challenges discussed in this section.

8.2 Mapping key challenges to the framework

This subsection builds upon the challenges defined in Sect. 4 and explains how the proposed multi-layer framework addresses each of them through coordinated cross-layer mechanisms enabled by the *Reaction Layer*.

CH1 – Data management strategy: To address the lack of a clear data management strategy, the framework includes lifecycle-aware access control, structured data classification, and policy-driven data handling mechanisms across the cloud and edge layers. The *Reaction Layer* continuously verifies and applies these policies, ensuring accountability, transparency, and adaptive risk management throughout the data lifecycle.

CH2 – Data governance: The framework operationalises governance principles by turning policy rules into enforceable technical controls. Role-based access management, audit logging, compliance monitoring, and dynamic policy updates are coordinated through the *Reaction Layer*, enabling governance policies to be enforced in practice rather than remaining conceptual.

CH3 – Integration and interoperability: To address interoperability and integration challenges, the framework uses standardised interfaces, secure APIs, semantic data harmonisation, and middleware coordination across the sensor, network, and edge layers. Context broker architectures are one established solution for this purpose [53]. Because such architectures achieve interoperability through translation at gateways rather than through convergence on a shared protocol, every integration point is also a trust boundary. The

Reaction Layer therefore monitors cross-platform interactions and detects inconsistencies or anomalous integration behaviour, maintaining smooth system operation.

CH4 – Data security, privacy, and cyber threat impacts: Security mechanisms are built across all ADE layers. They include device authentication (at the perception layer), secure communication (at the network layer), anomaly detection (at the edge layer), and access control with encryption (at the cloud layer). The *Reaction Layer* correlates security events, evaluates threat severity, and organises coordinated prevention, detection, and mitigation actions so that incidents are contained before they propagate across layers.

CH5 – Knowledge sharing and inclusiveness: Although largely human-centred, this challenge is technically supported by feedback mechanisms, transparent policy visualisation, and incident reporting dashboards coordinated at the cloud layer. The *Reaction Layer* ensures the consistent distribution of security insights and operational feedback across stakeholders, promoting trust and informed participation.

CH6 – Equipment and infrastructure constraints: To address infrastructure limitations, the framework incorporates adaptive deployment strategies, edge-based local processing, and failover mechanisms, which keep the system available during connectivity disruptions. The *Reaction Layer* adjusts its security configuration and enforcement rules according to the prevailing operating conditions, allowing it to function under constrained resources.

By mapping each challenge to concrete architectural mechanisms, the proposed framework moves beyond simple problem identification. It introduces an operational structure that supports coordinated, adaptive, and cross-layer Cybersecurity management within ADEs.

8.3 The reaction layer cross-layer coordination algorithm

To operationalize the *Reaction layer* described in Section 8, this subsection presents the *Reaction Layer Cross-Layer Coordination* (RLCC) algorithm whose full pseudocode is given in Algorithm 1. The RLCC formalizes the sequential logic by which the *Reaction Layer* collects security events, assesses threat severity, and deploys coordinated countermeasures across the ADE layers. Following the Prevention–Detection–Mitigation (PDM) security model [25], the algorithm is structured into four phases. Unlike existing approaches reviewed in Table 1, which address individual layers in isolation, the RLCC explicitly encodes cross-layer coordination as a first-class architectural concern, with Phase 3 constituting the core technical contribution.

Algorithm 1: Reaction Layer Cross-Layer Coordination (RLCC)

Input: $\mathcal{E}$: events from $\mathcal{L} = \{P, N, E, C\}$; $\mathcal{T}$: STRIDE taxonomy; $\mathcal{V}$: system state vector; Θ : policy base (GDPR, NIS2, MITRE ATT&CK); θ_{lo}, θ_{hi} : severity thresholds; r_{max} : escalation bound

Output: $\mathcal{A}$: response actions; $\mathcal{CH}_{res}$: resolved challenges; Θ' : updated policy base

```

/* Phase 1 — Prevention */
1  $r \leftarrow 0$ ;  $\Theta' \leftarrow \Theta$ ; Enforce  $\Theta$  on all  $\mathcal{L}$ ; disseminate security policies to stakeholders; if  $\mathcal{E} = \emptyset$  then remain in Phase 1 */
/* Phase 2 — Detection */
2 foreach  $e \in \mathcal{E}$  do classify  $l_e, t_e \in \mathcal{T}, \sigma_e \in [0, 1]$  */
3 Compute  $\sigma_{max}$ ; correlate events; propagate alerts to  $\mathcal{L}_{aff}$ ; if  $\sigma_{max} < \theta_{lo}$  then goto Phase 1
4 if  $\sigma_{max} \geq \theta_{hi}$  then  $\mathcal{L}_{aff} \leftarrow \mathcal{L}$  ▷ ecosystem-wide response
/* Phase 3 — Mitigation: Cross-Layer Actuation */
5  $\mathcal{A} \leftarrow \emptyset$ ;  $\mathcal{CH}_{res} \leftarrow \emptyset$  */
6 if  $P \in \mathcal{L}_{aff}$  then ▷ Perception Layer — Device Auth & Sensor Isolation
7    $\mathcal{D}_c \leftarrow \{d \in Dev(P) : AuthFail(d) \vee Anomaly(d)\}$ 
8   foreach  $d \in \mathcal{D}_c$  do RevokeCredentials( $d$ ); Blacklist( $d, \Theta$ ); if  $t_e \in \{Spoofing, Tampering\}$  then IsolateSensor( $d$ ); ActivateBackup( $d$ )
9
10  foreach  $d \in Dev(P) \setminus \mathcal{D}_c$  do if not Validate( $d, \mathcal{V}_P$ ) then Flag( $d$ )
11
12   $\mathcal{A} \leftarrow \mathcal{A} \cup \{(P, Auth\_Iso)\}$ ;  $\mathcal{CH}_{res} \leftarrow \mathcal{CH}_{res} \cup \{CH4, CH6\}$ 
13 end
14 if  $N \in \mathcal{L}_{aff}$  then ▷ Network Layer — Dynamic Access Control & Traffic Filtering
15   foreach  $f \in Flows(N)$  do if not Compliant( $f, \Theta$ ) then BlockFlow( $f$ ); UpdateACL( $f, \Theta$ )
16
17   foreach  $p \in Traffic(N)$  do if Malicious( $p, T$ ) then Drop( $p$ ); Log( $p, t_e$ )
18
19   foreach  $s \in Src(N)$  do if Rate( $s$ ) > Thresh( $\Theta$ ) then RateLimit( $s$ )
20
21   foreach  $c \in API(N)$  do if not Auth( $c, \Theta$ ) then Reject( $c$ ); Log( $c$ )
22
23    $\mathcal{A} \leftarrow \mathcal{A} \cup \{(N, ACL\_Filter)\}$ ;  $\mathcal{CH}_{res} \leftarrow \mathcal{CH}_{res} \cup \{CH3, CH4\}$ 
24 end
25 if  $E \in \mathcal{L}_{aff}$  then ▷ Edge Layer — Containment & Workload Isolation
26    $\mathcal{Q}_c \leftarrow \{n \in Nodes(E) : IntViol(n) \vee ResAnomaly(n)\}$ 
27   foreach  $n \in \mathcal{Q}_c$  do Quarantine( $n$ ); Redirect( $n, SafeNode$ )
28   foreach  $w \in Work(E)$  do if Affected( $w, \mathcal{Q}_c$ ) then Suspend( $w$ ); Migrate( $w$ )
29
30   foreach  $s \in Svc(E)$  do if Degraded( $s$ ) then Failover( $s$ ); NotifyCloud( $s, C$ )
31
32    $\mathcal{A} \leftarrow \mathcal{A} \cup \{(E, Contain\_Iso)\}$ ;  $\mathcal{CH}_{res} \leftarrow \mathcal{CH}_{res} \cup \{CH4, CH6\}$ 
33 end
34 if  $C \in \mathcal{L}_{aff}$  then ▷ Cloud/Application Layer — Access Revocation & Data Recovery
35   foreach  $k \in Tok(C)$  do if Compromised( $k$ )  $\vee$  PolViol( $k, \Theta$ ) then Revoke( $k$ ); Notify( $k$ ) ▷ CH2
36
37   foreach  $d \in Data(C)$  do if IntViol( $d$ ) then Rollback( $d$ ); Validate( $d$ ) ▷ CH1
38
39   foreach  $p \in \Theta$  do if Breach( $p, t_e$ ) then  $\Theta \leftarrow \Theta \setminus \{p\} \cup Upd(p, t_e, \mathcal{V})$ ; Enforce( $\Theta, C$ ) ▷ CH2
40
41   StoreLog( $\mathcal{E}, \mathcal{A}, \sigma_{max}, \mathcal{CH}_{res}, \Theta$ ) ▷ GDPR, NIS2 — CH2
42    $\mathcal{A} \leftarrow \mathcal{A} \cup \{(C, Revoke\_Recover)\}$ ;  $\mathcal{CH}_{res} \leftarrow \mathcal{CH}_{res} \cup \{CH1, CH2\}$ 
43 end
44  $\mathcal{V} \leftarrow \mathcal{V} \oplus StateChange(\mathcal{A})$ 
/* Phase 4 — Audit & Feedback */
45 Disseminate security report to stakeholders ▷ CH5
46 if ThreatContained( $\mathcal{V}$ ) then
47   RestoreServices( $\mathcal{L}_{aff}$ );  $\Theta' \leftarrow Recalibrate(\Theta, t_e, \mathcal{A}, \mathcal{V})$ ; goto Phase 1 with  $\Theta'$ 
48 else
49   Escalate( $\mathcal{E}, \mathcal{A}, \sigma_{max}$ );  $r \leftarrow r + 1$ ; if  $r \leq r_{max}$  then goto Phase 3 else hand off to administrator
50 end
51 return  $\mathcal{A}, \mathcal{CH}_{res}, \Theta'$ 

```

Phase 1 — Prevention.

Prevention is the default operating state of the *Reaction Layer*. During this phase, the policy base Θ is enforced across all ADE layers. It includes security and governance rules from GDPR, NIS2 (Network and Information Security Directive), and MITRE ATT&CK (Adversarial Tactics, Techniques, and Common Knowledge). Each ADE layer applies the policies according to its role. The Perception layer checks device authentication and user credentials. The Network layer enforces access control lists and API authorization rules. The Edge layer monitors workload integrity. The Cloud layer applies data governance and data lifecycle policies. The Reaction Layer continuously monitors all layers for security events $\mathcal{E}$. If $\mathcal{E} = \emptyset$, the system stays in Prevention mode. This phase mainly addresses CH1 and CH2. It applies governance rules and security policies before an attack occurs. This proactive approach is not provided by the conceptual governance models reviewed in Sect. 5.

Phase 2 — Detection.

When a security event $e \in \mathcal{E}$ is detected, the Reaction Layer starts the detection process. First, the system identifies the source layer l_e . Next, it maps the event to a STRIDE threat category $t_e \in \mathcal{T}$. Then, it calculates a severity score $\sigma_e \in [0, 1]$. The score is based on the current system state $\mathcal{V}$, the impact on the five security goals shown in Table 3, and the possibility that the attack can spread across different ADE layers. The *Reaction Layer* also checks whether similar events occur in multiple layers during the same time window. If this happens, the severity score is increased because coordinated attacks are more dangerous than isolated events. After all events are analyzed, the system finds the highest severity score $\sigma_{\max}$. It also identifies the affected layers $\mathcal{L}_{\text{aff}} \subseteq \mathcal{L}$. The next action depends on two thresholds, $0 < \theta_{lo} < \theta_{hi} \leq 1$. If $\sigma_{\max} < \theta_{lo}$, the event is recorded, and the system returns to Phase 1. If $\theta_{lo} \leq \sigma_{\max} < \theta_{hi}$, the event is sent to Phase 3. Only the affected layers $\mathcal{L}_{\text{aff}}$ perform the mitigation actions.

If $\sigma_{\max} \geq \theta_{hi}$, the incident is treated as a system-wide attack. All ADE layers are included in the response, and Phase 3 starts with full mitigation. The lower threshold θ_{lo} helps reduce unnecessary alerts. The higher threshold θ_{hi} ensures that high-severity or coordinated multi-layer attacks always receive a fast and coordinated response across the entire ADE. The formal definitions of σ_e , θ_{lo} , and θ_{hi} are presented in Sect. 8.4.

Phase 3 — Mitigation: cross-layer actuation.

Phase 3 is the main part of the RLCC. At the beginning, the response action set $\mathcal{A}$ and the resolved challenge set $\mathcal{CH}_{\text{res}}$ are initialized as empty. Then, the system starts one mitigation procedure for each affected layer in $\mathcal{L}_{\text{aff}}$. Instead of using one response for all layers, the *Reaction Layer* applies four layer-specific procedures. Each procedure supports the next one and helps reduce the remaining attack surface.

At the Perception layer, the system identifies compromised devices $\mathcal{D}_c = \{d \in \text{Dev}(P) : \text{AuthFail}(d) \vee \text{Anomaly}(d)\}$. Their credentials are revoked, and the devices are added to a blacklist according to Θ . Sensors showing Spoofing or Tampering behavior are isolated, and backup sensors are activated to keep data collection running. This helps prevent infrastructure limitations from causing service failures (CH6). The remaining devices are checked against $\mathcal{V}_P$, and suspicious readings are marked for further review.

At the Network layer, the system performs four actions. First, it updates access control rules to block traffic that violates the active security policy Θ . Second, it inspects incoming traffic for patterns related to the detected threat t_e . Suspicious packets are dropped and logged. Third, traffic that exceeds the allowed rate $\text{RateThreshold}(\Theta)$ is limited instead of being completely blocked. This reduces flooding attacks while allowing normal traffic to continue. Fourth, unauthorized API requests are rejected and recorded. These actions help solve CH3 by enforcing consistent security rules and CH4 by stopping attacks before they spread to the Edge or Cloud layers.

At the Edge layer, compromised nodes $\mathcal{E}_c = \{n : \text{IntViol}(n) \vee \text{ResAnomaly}(n)\}$ are isolated. Their workloads are moved to trusted nodes. Affected workloads are suspended and transferred to isolated environments to stop lateral movement. If needed, failover services are activated. These actions support CH4 by containing attacks and CH6 by maintaining system operation under limited resources.

At the Cloud/Application layer, the system performs four actions. First, it revokes the access tokens of compromised or unauthorized users and informs the affected stakeholders. This supports the governance requirements (CH2). Second, data with compromised integrity are restored from the latest verified backup and checked again before use (CH1). Third, security policies that failed to stop the detected threat are updated as $\Theta \leftarrow \Theta \setminus \{p\} \cup \text{Upd}(p, t_e, \mathcal{V})$ and applied immediately. Fourth, the system stores a complete incident record, including detected events, response actions, and resolved incidents. This record supports the GDPR and NIS2 compliance (CH2).

After all mitigation actions are completed, the system updates the global state:

$$\mathcal{V} \leftarrow \mathcal{V} \oplus \text{StateChange}(\mathcal{A})$$

This update reflects the changes made by deployed countermeasures across the affected layers.

Phase 4 — Audit and feedback.

The *Reaction Layer* creates a security report and shares it with the relevant stakeholders. This improves transparency and addresses CH5. If $\text{ThreatContained}(\mathcal{V})$ returns true, the affected services are restored, and the policy base is updated:

$$\Theta' \leftarrow \text{Recalibrate}(\Theta, t_e, \mathcal{A}, \mathcal{V})$$

The updated policy base is used in the next Prevention cycle. This reduces the chance that the same attack will succeed again. The system then returns to Phase 1, completing the feedback loop shown in Fig. 3. If the threat is not contained, the incident is escalated. The escalation counter r is increased, and the system returns to Phase 3 with the updated system state. To ensure that the process always finishes, the number of retries is limited by $r_{\max}$. After $r_{\max}$ unsuccessful attempts, the system hands control to the administrator. This allows a human expert to handle new or persistent attacks. By using four coordinated mitigation procedures, the RLCC ensures that all ADE layers work together. Each procedure is linked to one or more challenges (CH1–CH6) described in Sect. 4. This provides coordinated security, governance support, and cross-layer protection that are not available in the existing approaches reviewed in Sect. 2.

8.4 Formal properties and positioning of the RLCC algorithm

This subsection defines the main quantities used in Algorithm 1. It then explains the algorithm's termination and safety properties. Next, it analyses the computational complexity. Finally, it explains how the proposed approach relates to existing security orchestration technologies.

Severity scoring. For each event e , the severity score is computed as

$$\sigma_e = w_I I(e) + w_G G(e) + w_X X(e)$$

where the weights satisfy $w_I + w_G + w_X = 1$. The score is based on three factors: The first factor, $I(e)$, measures “how important the affected asset is”. Its value is taken from the system state vector $\mathcal{V}$. The second factor, $G(e)$, measures “how many of the five security goals in Table 3 are affected by the STRIDE threat type t_e ”. The third factor, $X(e)$, measures “how easily the attack can spread across ADE layers”. It is calculated as

$$(\lambda(e) - 1) / (|\mathcal{L}| - 1),$$

where $\lambda(e)$ is the number of ADE layers that can be reached from the source layer l_e (potential propagation). The subtraction of 1 excludes the source layer from the calculation. Thus, $X(e)=0$ when an attack remains within its source layer, and $X(e)=1$ when it can propagate to all other ADE layers.

Phase 2 also increases the severity score when related events appear in multiple layers. If events of the same threat class are detected in $m \geq 2$ layers during the same correlation window, their severity becomes

$$\sigma_e \leftarrow \min\left(1, \sigma_e + \beta \frac{m-1}{|\mathcal{L}|-1}\right),$$

where m is the number of layers that have actually observed propagation and β controls the effect of cross-layer correlation. As a reference configuration, we use $w_I = w_G = w_X = 1/3$ and $\beta = 0.25$. These values are only design defaults. They will be calibrated experimentally in the prototype described in Sect. 9. A learned severity model is left as future work (Section 10).

Threshold parameterization. The thresholds satisfy

$$0 < \theta_{lo} < \theta_{hi} \leq 1.$$

They divide the severity score into three levels:

- If $\sigma_{\max} < \theta_{lo}$, the event is only logged.
- If $\theta_{lo} \leq \sigma_{\max} < \theta_{hi}$, targeted mitigation is applied to the affected layers $\mathcal{L}_{\text{aff}}$.
- If $\sigma_{\max} \geq \theta_{hi}$, the response is extended to the whole ADE.

The thresholds balance two objectives. A low threshold may create too many alerts, whereas a high threshold may miss serious attacks. As reference values, we propose $\theta_{lo} = 0.3$, $\theta_{hi} = 0.7$, and $r_{\max} = 3$. $r_{\max}$ is the maximum number of mitigation attempts before manual intervention is required. These values are deployment parameters and should be adjusted during system validation.

Policy recalibration. The recalibration function is defined as:

$$\text{Recalibrate} : (\Theta, t_e, \mathcal{A}, \mathcal{V}) \rightarrow \Theta'.$$

where Θ is the current set of security and policy rules, t_e is the STRIDE threat type of the detected event, $\mathcal{A}$ is the mitigation actions applied during the incident, and $\mathcal{V}$ is the current system state, including asset status and security conditions. Recalibrate updates the policy base after each incident as follows:

$$\Theta' = (\Theta \setminus \Theta_b) \cup \{\text{Upd}(p, t_e, \mathcal{V}) : p \in \Theta_b\} \cup \text{Der}(t_e, \mathcal{A}),$$

where

$$\Theta_b = \{p \in \Theta : \text{Breach}(p, t_e)\}.$$

Θ_b contains the policies that were not sufficient to stop the attack. A policy p is considered breached if it fails to prevent or mitigate the detected threat t_e . The function *Upd* strengthens these policies, for example by reducing rate limits, changing ACL (Access Control List) rules, or shortening token lifetimes. The function *Der* creates new preventive

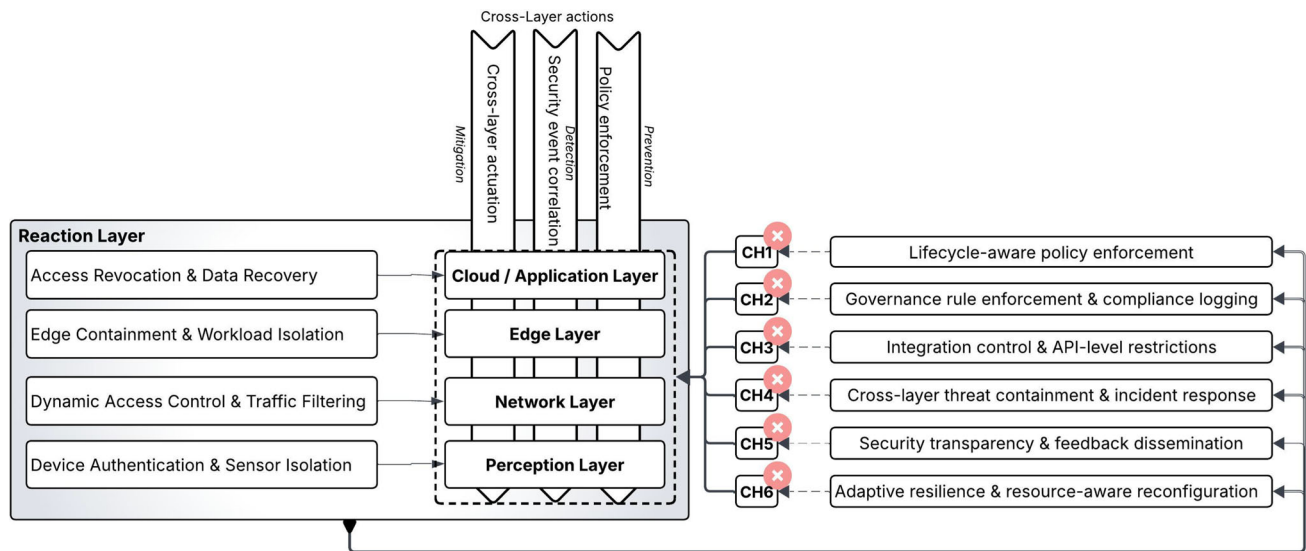


Fig. 3 Conceptual framework of a secure agricultural data-sharing ecosystem

rules from the successful mitigation actions. The recalibration process must always preserve regulatory compliance. Rules required by GDPR and NIS2 cannot be removed. Only stronger rules or additional rules can be added. Therefore, if Θ satisfies the required regulations, Θ' also satisfies them.

Termination and safety. Each execution of the RLCC algorithm always finishes. The loops in Phases 2 and 3 process finite sets of entities, such as devices, services, flows, workloads, and policies. Therefore, each phase finishes after a finite number of steps. The only repeated execution is the transition from Phase 4 back to Phase 3. This repetition is limited by the escalation counter $r \leq r_{\max}$. After the maximum number of attempts, the control is transferred to the system administrator. As a result, Phase 3 is executed at most $1 + r_{\max}$ times. The algorithm also satisfies a safety property. After Phase 3 finishes, every entity that violates the active policy base Θ has been processed. The mitigation actions are stored in $\mathcal{A}$, and every action is recorded in the compliance log. These are design-level arguments rather than formal proofs. Formal verification, for example by model checking, is left for future work.

Complexity. Let n be the total number of managed entities in the four ADE layers. Let $|\mathcal{E}|$ be the number of events processed during one activation. Phase 2 requires $O(|\mathcal{E}|)$ time for event classification and severity scoring. The worst-case correlation cost is $O(|\mathcal{E}|^2)$, because every event is compared with all other events. Using time-windowed correlation reduces this cost to $O(|\mathcal{E}| \log |\mathcal{E}|)$. This complexity assumes that events are sorted by timestamp using a standard sorting algorithm. Each execution of Phase 3 requires $O(n + |\Theta|)$. Therefore, one activation requires:

$$O((1 + r_{\max})(n + |\Theta|))$$

Phase 3 is executed once when the threat is first detected and may be repeated at most $r_{\max}$ additional times. The memory complexity is:

$$O(|\mathcal{E}| + |\mathcal{A}| + |\Theta|)$$

The mitigation cost therefore grows approximately linearly with the number of managed entities. This property is important for resource-constrained ADE environments (CH6).

8.5 Positioning with respect to cross-layer orchestration approaches

Automated and closed-loop security orchestration is not unique to the RLCC. Similar approaches are used in enterprise IT, telecommunications, smart grids, IoT, critical infrastructures, and autonomic computing. The closest comparison is with SIEM and SOAR platforms, with which the RLCC is designed to work rather than to replace: Phase 2 performs functions similar to SIEM event collection and correlation, and the mitigation actions of Phase 3 resemble the automated playbooks used by SOAR platforms. One difference is nevertheless decisive. SIEM and SOAR apply existing policies or reconfigure systems during operation [20, 43], but they do not update the policy base after an incident. The RLCC closes this loop through the *Recalibrate* function (Sect. 8.4), which strengthens the policies that failed during the incident and applies them in the next prevention cycle.

This subsection compares the RLCC with representative orchestration approaches from the domains listed above, and with the Agriculture 4.0 approaches of Table 1. Each approach was assessed from its published specification or peer-reviewed description, since the internal orchestration

logic of commercial products is generally not disclosed; commercial SIEM and SOAR platforms are therefore represented as a system class rather than individually.

The comparison uses four properties. They reflect the structural challenges identified in Sect. 4, where the recurring deficiency was not the absence of individual security controls but the absence of a mechanism connecting governance rules, technical enforcement, and incident response across layers. **P1 (cross-layer scope)** measures whether one control loop manages the perception, network, edge, and cloud layers. **P2 (formal decision logic)** measures whether the response process is defined as an algorithm with clear inputs, states, and stopping conditions. **P3 (closed-loop policy recalibration)** measures whether security policies are updated automatically after an incident. **P4 (regulatory traceability)** measures whether each response action is linked to compliance requirements and recorded in an auditable log. Deployment conditions, which constrain where a system can operate rather than what it does, are not scored as properties; they are reported separately in the final column of Table 7.

Each property was scored using the documented capabilities of each system. A property is marked as satisfied when the source clearly describes a mechanism for it, as partially satisfied when the mechanism is limited in scope or only partly specified, and as not satisfied when no such mechanism is described. Table 7 summarises the comparison.

Several approaches satisfy some properties, and a few satisfy most. Cross-vendor standards define some parts of the response process but do not provide a complete solution. CACAO 2.0 defines a standard playbook workflow, but it does not specify how to determine the outcome of each step or how the workflow should be executed [2]. OpenC2 standardises security commands, but it does not include a workflow or a decision-making model [63]. The ETSI ZSM framework supports closed-loop management across multiple domains but is not designed specifically for security at the reference architecture level [1]. The autonomic computing vision [46] introduced the closed-loop control idea but remains a reference architecture: it does not define the decision logic, the policy update process, or regulatory requirements, which the RLCC specifies for agricultural digital ecosystems. SOAR4IoT is closest for P2, defining a response process based on events, assets, and playbooks [33]; however, its playbooks are created manually, their selection depends on a security analyst, and automatic generation is left for future work, so P3 remains unsupported because incidents do not update the policy base. In telecommunications, INSPIRE-5Gplus is closest overall, instantiating the ZSM closed loop with explicit security modules in each management domain and at the end-to-end level, where its decision engine and security orchestrator allocate, chain and reconfigure virtual security functions and control traffic through SDN controllers [57]. Its policies are Security Service Level

Agreements specified by verticals and end users: on violation the platform selects and enforces the most suitable available policy but does not derive new policy parameters from the incident, so P3 is partially satisfied; and its decision logic is assigned to the Security Intelligence Engine rather than specified as an algorithm with defined states and stopping conditions, so P2 is also partially satisfied. In agriculture, by contrast, mitigation measures are presented as per-layer recommendations and standards-based guidance rather than as an orchestrated response process, and compliance is treated as adherence to external standards rather than as traceability of individual response actions [50].

The main difference is the deployment environment rather than the control mechanism. Several approaches involve multiple stakeholders, such as smart grids with generators, network operators and prosumers [39], and CyberSANE with information sharing between organisations [60], but they still operate under one security authority. INSPIRE-5Gplus is developed for single-operator environments and extends to other operators and OTT providers through the cross-domain integration fabric, so multi-party interaction is expressed through agreements between operators rather than between independent asset owners [57]. In an ADE, by contrast, farmers, equipment vendors, cooperatives and processors each own their data and control their own responses, so no single organisation controls the whole response process. Existing systems also assume reliable networks and sufficient computing resources. INSPIRE-5Gplus assumes a persistent integration fabric, ledger nodes that participate continuously in validation, and trusted execution and attestation hardware. SOAR4IoT supports constrained devices by hosting the decision logic in digital twins on external servers, which requires continuous synchronisation and is difficult under the intermittent connectivity of agricultural environments (CH6).

Most existing systems support compliance through logging or reporting after an incident. INSPIRE-5Gplus is the strongest exception, binding component obligations to regulatory, vertical and standards requirements through manifests, expressing required security capabilities using an established control framework, and recording message interactions in permissioned distributed ledgers to support non-repudiation and attribution [57]. This binding is nevertheless defined at component and agreement level rather than at the level of each response action. The RLCC instead applies compliance rules during policy recalibration and links every response action to CH1–CH6. This also improves robustness. Kephart and Chess [46] explain that changing security policies creates a new attack surface, because an attacker may try to modify high-level policies. The monotonicity constraint on Θ' (Sect. 8.4) addresses this risk, since recalibration can strengthen security rules but cannot remove controls required by GDPR or NIS2.

Table 7 Positioning of the RLCC against cross-layer security orchestration approaches in adjacent domains. ✓ = satisfied; ~ = partially satisfied; × = not satisfied

Approach / system class	Primary domain	P1	P2	P3	P4	Deployment assumption
Security orchestration platforms (SIEM/SOAR)	Enterprise IT	~	×	×	~	Single organisational domain; response logic manually configured; behaviour under intermittent connectivity varies by product
Playbook workflow standards (CACAO 2.0)	Cross-vendor IT	~	~	×	×	Normative workflow model; step outcome determination, transport and execution engine left to implementations
Command and control standards (OpenC2)	Cross-vendor IT	~	×	×	×	Interoperable action semantics at the command layer; no workflow or decision model
Closed-loop network automation (ETSI ZSM)	Telecommunications	✓	~	✓	~	Multiple domains under one framework owner; assumes a managed integration fabric and persistent data services; security-agnostic at reference-architecture level
AI-driven zero-touch 5G security orchestration (INSPIRE-5Gplus)	Telecommunications (5G/B5G)	✓	~	~	~	Single-operator focus, extended to other operators and OTT providers through the cross-domain integration fabric; assumes a persistent integration fabric, permissioned-ledger stewards participating continuously in validation, and TEE/TPM-capable infrastructure
SDN-based grid security orchestration	Smart grid	~	×	×	~	Single-domain utility deployment across the TSO/DSO/DER chain; centralised SDN control; requires OpenFlow-capable equipment and reliable in-domain connectivity

Table 7 continued

Approach / system class	Primary domain	P1	P2	P3	P4	Deployment assumption
IoT security orchestration with digital twins (SOAR4IoT)	IoT / IIoT	~	~	×	×	Single domain with security analysts; device and network layers only; replication-mode twins require continuous bidirectional synchronisation; logic hosted off-device
Collaborative CII incident handling (Cyber-SANE)	CII (health, transport, energy)	~	~	~	~	Institutional operators with a response team; mitigation proposed for human selection; regulation constrains data handling, not response actions
Autonomic (MAPE-K) self-managing architectures	Domain-agnostic	~	~	~	×	Reference loop over a single managed element; system-scale behaviour emerges from negotiation; policies supplied by humans; regulatory mechanisms left open
Agriculture 4.0 approaches (Table 1)	Agriculture	×	×	×	×	Single-layer or conceptual; no orchestration mechanism
RLCC (this study)	Agriculture (ADE)	✓	✓	✓	✓	Multi-stakeholder, resource-constrained, intermittently connected

This paper therefore does not claim that no similar orchestration mechanism exists. It claims that none of the reviewed approaches satisfies all four properties at the same time under the multi-stakeholder, resource-constrained and intermittently connected conditions of an ADE. Two limitations apply. The comparison is based on documented capabilities, so undocumented features of commercial platforms may be missing, and it is representative rather than exhaustive.

9 Towards an initial prototype

This section describes the architecture of an initial RLCC prototype. The goal is not to define a complete implementation. Instead, it identifies which parts of the RLCC can be built with existing technologies and which parts require new development. This distinction shows what the framework reuses and what it contributes. It also addresses the practical feasibility of deploying the RLCC.

9.1 Prototype architecture

The design of this prototype has several constraints. Kampourakis et al. reviewed 46 wireless security testbeds used in water, healthcare, transportation, energy, maritime, and agricultural critical infrastructures [44]. They found that most testbeds have a narrow scope and are difficult to reuse. They also found that there is no common reference architecture for designing testbeds across different sectors. In addition, none of the reviewed testbeds produced publicly available datasets. The study also shows that security in cyber-physical systems requires a trade-off. Security controls consume bandwidth, energy, and increase latency. However, many availability-critical systems have limited resources. The same limitation applies to the RLCC coordination service. Its overhead must remain within the resource constraints described in CH6. For this reason, Table 9 includes resource-footprint and scalability as evaluation metrics.

The Perception, Network, Edge, and Cloud layers can be partly implemented using existing technologies. The Perception layer can be emulated using IoT simulation platforms that support device authentication, sensor scripting, and attack injection. The Network layer can be implemented using Software-Defined Networking (SDN), which supports dynamic ACL updates and traffic filtering. The Edge layer can use container platforms that already support quarantine and workload migration. The Cloud/Application layer can use open-source cloud management platforms, publish/subscribe context brokers, policy engines, logging frameworks, and compliance tools. Publish/subscribe context brokers provide a practical way to transfer security events across these layers. Martínez et al. developed a FIWARE-based farm management system in which gate-

way protocol adapters convert different device protocols into a common context format. A chain of brokers then separates context producers from context consumers [53]. This separation is important for the RLCC because the *Reaction Layer* must receive security events from all ADE layers without depending on the devices or services that generate them.

The main component that cannot be reused from existing technologies is the *Reaction Layer* coordination service. This service is the main architectural contribution of the RLCC. It subscribes to security events from all four ADE layers. It computes the severity score σ_e using the system state vector $\mathcal{V}$. It identifies the affected layers $\mathcal{L}_{\text{aff}} \subseteq \mathcal{L}$ and sends the appropriate mitigation actions to each layer.

Existing SIEM and SOAR platforms provide only partial support, as discussed in Sect. 8.5. The severity thresholds θ_{lo} and θ_{hi} and the policy update function $\text{Upd}(p, t_e, \mathcal{V})$ have no direct equivalent in the approaches reviewed. These components must therefore be implemented specifically for the RLCC. The remaining components can be adapted from existing technologies. Table 8 summarises this mapping.

9.2 Validation scenarios and evaluation metrics

To validate the prototype, three representative threat scenarios drawn from the STRIDE-based threat model (Sect. 8) would be exercised. The first is sensor spoofing at the Perception layer, which would trigger credential revocation and backup sensor activation. The second is DDoS flooding at the Network layer, triggering rate limiting and cross-layer alert propagation. The third is ransomware-style data tampering at the Edge layer, triggering node isolation, workload migration, and notification of the Cloud layer. These three were selected because each originates at a different layer, so that the cross-layer propagation behaviour of the RLCC can be observed from three distinct starting points. Each scenario would be executed against a baseline of normal ADE traffic, so that the false-positive rate has a defined denominator.

Each scenario would be evaluated using the metrics in Table 9, grouped into four categories: latency, accuracy, scalability, and compliance. The accuracy metrics calibrate the thresholds θ_{lo} and θ_{hi} and the severity weights, for which Sect. 8.4 provides only initial values. The scalability metrics evaluate the complexity analysis presented in the same section, and the compliance metrics evaluate properties P3 and P4 from Sect. 8.5. Two metrics are of particular importance: recalibration effectiveness tests the closed-loop property that distinguishes the RLCC from existing orchestration platforms, since it measures whether a repeated attack is prevented at Phase 1 rather than handled again at Phase 3; and resource footprint tests deployability under the con-

Table 8 Mapping of RLCC prototype components to existing technologies

Component	Existing technologies	Role in prototype (aligned with RLCC phases)
Perception layer	IoT simulators, digital twins, adversarial injection frameworks	Supports Phase 1 (Prevention) via device authentication and Phase 2 (Detection) via spoofing and anomaly event generation
Network layer	SDN (Software-Defined Networking) controllers, ACL engines, traffic inspection tools	Enables Phase 1 policy enforcement and Phase 3 mitigation via rate limiting, traffic filtering, and API access control
Edge layer	Kubernetes, Docker, orchestration platforms, workload migration tools	Executes Phase 3 mitigation via node quarantine, workload migration, and failover activation
Cloud / Application Layer	Cloud management platforms, publish/subscribe context brokers, policy engines, logging frameworks, compliance tools (GDPR/NIS2)	Supports Phase 1 governance enforcement, Phase 2 event aggregation, Phase 3 token revocation and data recovery, and Phase 4 audit logging and stakeholder reporting
Reaction layer (Core RLCC Engine)	Partial overlap with SIEM/SOAR platforms and closed-loop orchestration architectures; see Table 7	Orchestrates Phases 2–4: security event ingestion, severity scoring, affected layer selection, and layer-specific mitigation dispatch

straints of CH6, which the testbed literature identifies as the binding limitation in cyber-physical settings. No target values are defined, since suitable thresholds depend on the deployment environment and should be determined during system calibration.

Existing measurements provide useful guidance for the event-processing path. Martínez et al. measured throughput and response time for different payload sizes and different numbers of devices transmitting concurrently [53]. Devices that send data asynchronously and disconnect to conserve energy achieved substantially lower throughput than devices holding the connection open, and performance degraded further as the number of concurrent devices increased. Their hardware and software are now dated and their devices were simulated rather than deployed, so the absolute values cannot be transferred. The trend, however, remains informative: it suggests that the event-processing path from the Perception layer, rather than broker capacity, is the more likely constraint on system performance. This supports the scalability metric of Table 9, and supports treating the number of managed entities n and the event rate $|\mathcal{E}|$ as separate variables, since each is bounded by a different scaling strategy.

Future work will implement the prototype and evaluate it against these metrics, as discussed in Sect. 10.

10 Future research directions

This study opens several directions for future research. The main priority is the empirical validation of the proposed framework through the prototype testbed outlined in Sect. 9. Specifically, the three threat scenarios—sensor spoofing at the Perception layer, DDoS flooding at the Network layer, and ransomware-like data tampering at the Edge layer—should be executed against a deployed instance of the RLCC coordination service and evaluated against the quantitative targets defined in Table 9. This empirical step would move the RLCC algorithm from a formally specified pseudocode to a tested operational component that closes the gap between conceptual design and practical deployment.

Second, the RLCC algorithm can be enhanced by integrating CTI feeds and machine learning-based anomaly detection into the severity scoring function σ_e . Currently, σ_e is computed against the system state vector $\mathcal{V}$ using rule-based criteria. Replacing this with a learned model, such as trained on agricultural IoT traffic patterns and known attack signatures, would make the Detection phase adaptive and predictive rather than purely reactive. It strengthens the framework's response to zero-day threats and novel attack vectors targeting ADE environments.

Table 9 Quantitative evaluation targets for the RLCC prototype

Metric	Operational definition	Claim or parameter tested	Scenarios
<i>Latency</i>			
Detection latency	Time from the arrival of an event at the Reaction Layer until Phase 2 is completed, including computation of the maximum severity score $\sigma_{\max}$ and identification of the affected layers $\mathcal{L}_{\text{aff}}$	Feasibility of event classification and severity scoring at operational event rates	Spoofing, DDoS, Tampering
Mitigation latency	Time to complete the layer-specific sub-procedures for all affected layers $\mathcal{L}_{\text{aff}}$ after Phase 2	Feasibility of coordinated cross-layer actuation (Phase 3)	Spoofing, DDoS, Tampering
<i>Accuracy</i>			
False-positive rate	Proportion of benign events for which $\sigma_e \geq \theta_{lo}$ and Phase 3 is entered	Calibration of θ_{lo} and of the weights w_I, w_G, w_X in σ_e	Background traffic
False-negative rate	Proportion of injected attacks for which $\sigma_{\max} < \theta_{lo}$ and no mitigation is triggered	Calibration of θ_{lo} ; adequacy of the goal mapping $G(e)$	Spoofing, DDoS, Tampering
Escalation rate	Proportion of incidents requiring $r \geq 1$ returns to Phase 3, and proportion reaching $r_{\max}$ and handed to an administrator	Adequacy of the escalation bound $r_{\max}$	Spoofing, DDoS, Tampering
Over-escalation rate	Proportion of incidents for which $\sigma_{\max} \geq \theta_{hi}$ and ecosystem-wide mitigation is triggered, but containment would have been achieved by the layers in $\mathcal{L}_{\text{aff}}$ alone	Calibration of θ_{hi} ; adequacy of the correlation weight β	Spoofing, DDoS, Tampering
Containment effectiveness	Proportion of incidents resolved without propagation beyond the layers in $\mathcal{L}_{\text{aff}}$ identified at Phase 2	Effectiveness of cross-layer coordination; adequacy of $X(e)$ and of the correlation weight β	Spoofing, DDoS, Tampering
<i>Scalability</i>			
Scalability under load	Detection and mitigation latency measured as functions of the number of managed entities n , the policy base size $ \Theta $, and the event rate $ \mathcal{E} $	The complexity claim $O((1 + r_{\max})(n + \Theta))$ per activation (Sect. 8.4); ingestion capacity under asynchronous Perception-layer traffic	DDoS
Resource footprint	Processor and memory consumption of the Reaction Layer coordination service during sustained operation	Deployability under the resource constraints of CH6	All scenarios; background traffic
<i>Compliance</i>			
Recalibration effectiveness	Whether replaying an identical attack after the update $\Theta' \leftarrow \text{Recalibrate}(\Theta, t_e, \mathcal{A}, \mathcal{V})$ is prevented at Phase 1 rather than requiring Phase 3	The closed-loop policy recalibration property (P3, Sect. 8.5)	Spoofing, DDoS, Tampering
Compliance completeness	Proportion of incident logs containing all fields required for GDPR and NIS2 reporting, as stored by <code>StoreLog</code> (Algorithm 1)	Regulatory traceability property (P4, Sect. 8.5)	Spoofing, DDoS, Tampering

Scenarios: Spoofing at the Perception layer; DDoS at the Network layer; Tampering at the Edge layer

Third, the governance, legal, and ethical implications of automated cross-layer security enforcement require dedicated investigation. The RLCC algorithm makes real-time decisions, like revoking device credentials, isolating edge nodes, and recalibrating governance policies without human intervention. The conditions under which such automated actions satisfy GDPR accountability requirements, NIS2 incident reporting obligations, and emerging AI governance frameworks remain an open research question that must be resolved before the framework can be deployed in regulated agricultural data-sharing environments.

Fourth, the framework's feasibility should be assessed in the context of emerging network infrastructures, including non-terrestrial networks (NTN), and 6G connectivity. These technologies will significantly change the communication and security landscape of Agriculture 4.0. The RLCC algorithm's functions and thresholds will need to be reviewed and updated to remain effective under next-generation network conditions.

Fifth, future developments in AI, especially generative AI and large language models (LLMs), are expected to change agricultural cybersecurity [37]. These technologies can support both attackers and defenders. On the defensive side, LLMs can improve the *Reaction Layer*. They can analyse security events from different ADE layers and support CTI. They can also summarise security data and explain automated mitigation decisions in a way that is easy for analysts to understand. This can improve the transparency required by CH5. On the offensive side, generative AI can make phishing and social engineering attacks easier to create. It can also help attackers find software vulnerabilities automatically [42]. In smart agriculture, farm operators with limited cybersecurity knowledge and resource-constrained IoT devices are likely targets. If the severity scoring function σ_e is replaced with a learned model, as proposed in this study, the framework will also face AI-specific threats. These include data poisoning, adversarial examples, and prompt injection. Therefore, the policy base Θ should include AI security and AI governance requirements, such as the EU AI Act. This will help the *Reaction Layer* monitor and protect the AI components used in the detection process.

Sixth, quantum computing is another technology that may affect the proposed framework. Recent studies identify quantum computing, together with AI and blockchain, as an important future technology for improving security in smart agriculture [50]. However, quantum computing also creates new security risks. Powerful quantum computers could break public-key cryptography, including RSA (Rivest–Shamir–Adleman) and elliptic-curve algorithms. These algorithms are currently used for device authentication, key exchange, and data protection across all ADE layers [55]. Agriculture is especially vulnerable to “harvest now, decrypt later” attacks. In this type of attack,

encrypted data are collected today and decrypted in the future when quantum computers become sufficiently powerful. This is a serious concern because agricultural records, such as land, yield, and financial data, are often stored for many years [55]. The move to post-quantum cryptography (PQC), such as the NIST-standardised ML-KEM (FIPS 203) [56], also creates challenges. Resource-constrained sensors in the Perception Layer may not be able to support the larger keys and ciphertexts used by PQC (Post-Quantum Cryptography). Therefore, crypto-agility should be an important requirement of the *Reaction Layer*. This will allow the cryptographic algorithms defined in Θ to be replaced or updated without changing the RLCC coordination logic.

Finally, the socio-economic dimensions of adoption are a remarkable barrier that researchers alone cannot address. For instance, the cost of the tools for small and rural farms, the digital literacy, and the incentive plans that would encourage farmers and stakeholders to share their data, require interdisciplinary collaboration between Cybersecurity researchers, agricultural economists, and policymakers. Addressing these aspects is essential because all related stakeholders ensure that the framework is technically robust, accessible, fair, and sustainable in Agriculture 4.0.

11 Conclusion

This study addressed the Cybersecurity challenges of Agriculture 4.0 by proposing a framework to protect the Agricultural Data-sharing Ecosystem (ADE) across the Perception, Network, Edge, and Cloud layers. It identified six main challenges: data management, data governance, integration and interoperability, data security and privacy, knowledge gaps, and infrastructure limitations (CH1–CH6).

The framework introduces a dedicated *Reaction Layer* to coordinate prevention, detection, and mitigation across all ADE layers. Its operation is defined by the Reaction Layer Cross-Layer Coordination (RLCC) algorithm. The algorithm specifies how prevention, detection, mitigation, and audit actions are performed after a threat is detected. It also links each response action to one or more of the six challenges. Most existing studies focus on protecting a single technology or a single layer. As a result, governance policies, security controls, and incident response remain separated. The main contribution of this study is not the protection of one layer. Instead, it provides a coordination architecture that brings these three elements together in one response process.

The study also presents an initial prototype architecture. It shows which components can be built using existing technologies and which require new development. It also defines the validation scenarios and evaluation metrics for the proposed framework. Future work will implement the prototype and evaluate it experimentally. It will also include a more

detailed risk analysis and deployment in real agricultural environments.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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